AAMC_MCAT_Exam_Review_Export_2026-06-25_18-25

Total captured: 59

Question 1 of 59

Data ID: 831341
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PASSAGE

Passage 1 (Questions 1-6)

About 15% of colorectal cancers (CRCs), or colon cancers, are caused by defects in the mismatch repair system (MMR) that corrects certain DNA replication errors. In some cases, this MMR defect contributes to microsatellite instability (MSI), which is characterized by genetic hypermutability in DNA microsatellites (small repeated sequences that occur throughout the genome).

Researchers studying MSI in CRC cell lines and primary tumors observed that 100% of those analyzed contained deletions of 3 to 8 base pairs in the 17-thymine nucleotide microsatellite (T_{17}) located in Intron 8 of the gene encoding heat shock protein 110 (HSP110). HSP110 can bind structurally similar heat shock proteins and functions to facilitate proper protein folding and to reduce levels of nonfunctional protein aggregates. The larger *HSP110* T_{17} deletions cause Exon 9 to be omitted from the final sequence during pre-mRNA processing. This generates a premature stop codon in Exon 10, resulting in the mutant protein HSP110ΔE9. People with MSI CRC have *HSP110ΔE9* transcripts in cancerous tissue only.

Table 1 shows, with respect to T_{17}-deletion size, the amount of *HSP110ΔE9* mRNA expressed relative to the total amount of *HSP110* mRNA expressed in MSI CRC primary tumor cells. The total amount of *HSP110* mRNA expressed includes both wild-type *HSP110* (HSP110WT) and *HSP110ΔE9* mRNA.

Table 1 Effect of T_{17}-deletion Size on Relative Expression of *HSP110ΔE9* in MSI CRC Tumor Cells

T_{17}-deletion size (bp)	*HSP110ΔE9* mRNA expressed (% of total *HSP110* mRNA expressed)
3	0
4	43
5	89
6	92
7	100
8	100

Table 2 summarizes how HSP110ΔE9 affects HSP110WT function in cells transfected with HSP110WT and/or HSP110ΔE9.

Table 2 Effects of HSP110ΔE9 on HSP110WT Function

Endogenous HSP110 form expressed	Heat shock-induced protein aggregation (% of total protein)	Cellular localization of HSP110 form(s)	Apoptotic cells after induced apoptosis (% of total cells)
none	100	na	nd
HSP110WT	64	nucleus, cytosol	25
HSP110ΔE9	96	cytosol	nd
HSP110WT + HSP110ΔE9	100	cytosol	50

(Note: na = not applicable; nd = not determined.)

Figure 1 shows the ability of human CRC cells expressing either HSP110ΔE9 or HSP110WT to produce tumors after injection into mice.

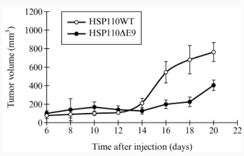

Figure 1 Effect of HSP110ΔE9 on tumorigenesis

QUESTION + ANSWER CHOICES

Based on the data presented in the passage, which statement best describes the *HSP110ΔE9* allele?

- A.
  Cancer-promoting and dominant to *HSP110WT*
- B.
  Cancer-promoting and recessive to *HSP110WT*
- **C.**
  Cancer-suppressing and dominant to *HSP110WT*
- D.
  Cancer-suppressing and recessive to *HSP110WT*

SOLUTION

Solution: The correct answer is **C**.

1. Based on Figure 1, expression of HSP110ΔE9 results in reduced, not increased, tumorigenicity. Thus, HSP110ΔE9 expression suppresses, it does not promote, cancer.
2. Based on Figure 1, HSP110ΔE9 expression suppresses, it does not promote, cancer. Furthermore, based on Table 2, HSP110ΔE9 expression is dominant, not recessive, to HSP110 because it counteracts HSP110Δ activity in protein aggregation.
3. **Figure 1 shows that HSP110ΔE9 expressing cells have a reduced tumor growth when injected in other mice. Based on Table 2, HSP110ΔE9 counteracts the effects of HSP110 on protein aggregation and stimulates apoptosis, thus HSP110ΔE9 expression is dominant to HSP110.**

4. Based on Figure 1, HSP110ΔE9 expression is cancer-suppressing; however, the effects of HSP110ΔE9 are dominant, and not recessive, to HSP110 because HSP110ΔE9 is able to counteract the effects of HSP110 and stimulate apoptosis.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
3 secs
Difficulty Level:
Content & Skills:
SIRS4
CC1B
FC1
BIO

Question 2 of 59

Data ID: 831348
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PASSAGE

Passage 1 (Questions 1-6)

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5	89
6	92
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8	100

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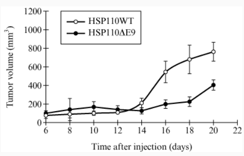

Figure 1 Effect of HSP110ΔE9 on tumorigenesis

QUESTION + ANSWER CHOICES

Based on the passage, in MSI CRC primary tumor cells, the size of the deletion in the *HSP110 T_{17}* microsatellite is inversely correlated with:

- A.
  the amount of HSP110ΔE9 protein expressed.
- **B.**
  the number of mature *HSP110WT* transcripts synthesized.
- C.
  the frequency of the omission of *HSP110* Exon 9 during splicing.
- D.
  the extent of premature translation termination in Exon 10.

SOLUTION

Solution: The correct answer is **B**.

1. The passage does not report protein levels for HSP110 or HSP110ΔE9.
2. **Table 1 shows that as the number of deleted base pairs (bp) increases from 3 to 8, the levels of HSP110ΔE9 increase whereas the levels of HSP110 decrease. This data indicates that there is an inverse relationship between the number of deleted bps and HSP110 levels.**
3. The passage does not show a frequency of E9 omissions. It only indicates that the larger omission results in deletion of E9.
4. The passage does not show the extent of premature translation termination, but only the levels of HSP110ΔE9 and HSP110 transcripts (mRNA).

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
2 mins 11 secs
Difficulty Level:
Content & Skills:
SIRS4
CC1B
FC1
BIO

Question 3 of 59

Data ID: 831357
Captured: 6/25/2026, 6:24:40 PM

PASSAGE

Passage 1 (Questions 1-6)

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Table 1 shows, with respect to T_{17}-deletion size, the amount of *HSP110ΔE9* mRNA expressed relative to the total amount of *HSP110* mRNA expressed in MSI CRC primary tumor cells. The total amount of *HSP110* mRNA expressed includes both wild-type *HSP110* (HSP110WT) and *HSP110ΔE9* mRNA.

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4	43
5	89
6	92
7	100
8	100

Table 2 summarizes how HSP110ΔE9 affects HSP110WT function in cells transfected with HSP110WT and/or HSP110ΔE9.

Table 2 Effects of HSP110ΔE9 on HSP110WT Function

Endogenous HSP110 form expressed	Heat shock-induced protein aggregation (% of total protein)	Cellular localization of HSP110 form(s)	Apoptotic cells after induced apoptosis (% of total cells)
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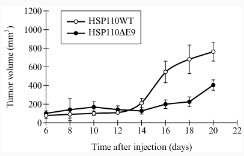

Figure 1 Effect of HSP110ΔE9 on tumorigenesis

QUESTION + ANSWER CHOICES

A man with a CRC mutation that results in the synthesis of HSP110ΔE9 and a woman that does not carry this mutation in any of her tissues have a child. What is the percent chance that the child will inherit the CRC mutation?

- **A.**
  0%
- B.
  25%
- C.
  50%
- D.
  100%

SOLUTION

Solution: The correct answer is **A**.

1. **Based on the passage, HSP110ΔE9 is only expressed in somatic cells, not germline. Thus, it cannot be inherited.**
2. The passage indicates that HSP110ΔE9 is only expressed in cancerous cells; therefore, it cannot be inherited.
3. As the transcripts are only expressed in cancerous cells, the mutation will not be inherited.
4. The child is not likely to inherit the mutation because, based on the passage, HSPΔE9 is only expressed in cancerous cells.

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RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
x
D
Time Spent:
2 mins 19 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1C
FC1
BIO

Question 4 of 59

Data ID: 831365
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PASSAGE

Passage 1 (Questions 1-6)

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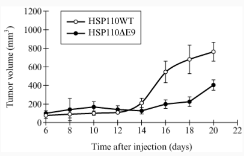

Figure 1 Effect of HSP110ΔE9 on tumorigenesis

QUESTION + ANSWER CHOICES

Which nucleotide pairing(s) would be recognized by the MMR system during DNA replication?

1. dTMP and dCMP
2. dGMP and dAMP
3. dAMP and dTMP

- A.
  I only
- B.
  III only
- **C.**
  I and II only
- D.
  I, II, and III

SOLUTION

Solution: The correct answer is **C**.

1. The MMR system recognizes DNA pair mismatches. dTMP-dCMP are a mismatch as well as dGMP and dAMP.
2. Option III is not a base mismatch, thus it will not be recognized by the MMR system.
3. **The passage indicates that MMR system recognizes the pair mismatch. Both options I and II represent mismatches, thus they will be recognized by the MMR system.**
4. Option I and II are mismatches and will be recognized by the MMR system. Option III is not a mismatch, and it will not be recognized.

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For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
x
C
D
Time Spent:
3 mins 23 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1B
FC1
BIO

Question 5 of 59

Data ID: 831367
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PASSAGE

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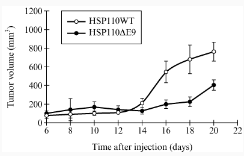

Figure 1 Effect of HSP110ΔE9 on tumorigenesis

QUESTION + ANSWER CHOICES

The information in the passage best supports the conclusion that Intron 8 of *HSP110* most likely contains which of the following?

- A.
  Stop codon
- **B.**
  Splice acceptor site
- C.
  HSP110 gene promoter
- D.
  Partial coding sequence of *HSP110*

SOLUTION

Solution: The correct answer is **B**.

1. Based on the passage, intron 8 does not contain a stop codon as translation ends with exon 10.
2. **It is most likely that intron 8 contains a splice acceptor site that allows exon 9 to be omitted upon mutation.**
3. The promoter is usually not located in an intron.
4. The intron does not code for a gene, an exon does.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer

A
B
C
x
D
Time Spent:
2 mins 0 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1B
FC1
BIO

Question 6 of 59

Data ID: 831370
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PASSAGE

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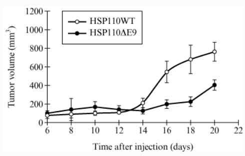

Figure 1 Effect of HSP110ΔE9 on tumorigenesis

QUESTION + ANSWER CHOICES

According to the information in the passage, HSP110 is most likely which type of protein?

- A.
  Adhesion
- **B.**
  Chaperone
- C.
  Clathrin
- D.
  Enzyme

SOLUTION

Solution: The correct answer is **B**.

1. The passage states that HSP110 is involved in protein folding. Adhesion proteins act in binding with other cells or with the extracellular matrix, not protein folding.
2. **The passage states that HSP110 is a heat shock protein, which facilitates proper protein folding and inhibits the formation of nonfunctional protein aggregates. Proteins that perform this function are chaperone proteins.**
3. Clathrin functions in formation of vesicles for intracellular trafficking, not in protein folding.
4. Based on the passage, HSP110 is involved in protein folding, not enzymatic catalysis of specific biochemical reactions.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct

Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
21 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1A
FC1
BCH

Question 7 of 59

Data ID: 831380
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PASSAGE

Passage 2 (Questions 7-11)

Malnutrition is known to increase susceptibility to colitis (colon inflammation). Colitis can also be caused by pellagra (a condition that may result from tryptophan deficiency) and Hartnup disease (a condition resulting from decreased function of the small intestinal neutral amino acid transporter AT1), the latter of which reduces gut absorption of tryptophan. The plasma membrane-embedded domain of the angiotensin I converting enzyme 2 (ACE2) is necessary for AT1 localization to the luminal plasma membrane of small intestinal epithelial cells. AT1 simultaneously transports tryptophan against its concentration gradient and a sodium ion along its concentration gradient into the cytoplasm of intestinal epithelial cells.

To determine how malnutrition promotes colitis, researchers treated wild-type (WT) mice and mice with loss-of-function mutations in the X-linked *Ace2* gene with dextran sodium sulfate (DSS), which increases colitis susceptibility by damaging intestinal epithelial cells. Mice were then fed one of the following three diets:

1. Standard diet (SD);
2. Protein-free diet (PFD); or
3. Gly–Trp (G–T), a dipeptide not requiring AT1 for intestinal uptake.

Mice were then monitored for weight loss as an indicator of colitis (Table 1).

Table 1 The Effects of *Ace2* Genotype and Diet on Colitis and Circulating Tryptophan

Ace2 genotype	Diet	Percent weight loss 4 days after start of DSS treatment (%)	Tryptophan blood levels (relative to control)
+/y (WT)	SD*	2	1.0
	PFD	22	not determined
	SD + G–T	2	1.4
–/y	SD*	10	0.3
	PFD	20	not determined
	SD + G–T	2	1.0

*Both mice genotypes on SD showed 0% weight loss without DSS.

QUESTION + ANSWER CHOICES

Which amino acid, in addition to tryptophan, is most likely to be transported by AT1?

- **A.**
  Phenylalanine
- B.
  Lysine
- C.
  Arginine
- D.
  Glutamate

SOLUTION

Solution: The correct answer is **A**.

1. **AT1 is a neutral amino acid transporter. Phenylalanine is a neutral amino acid.**
2. Based on the passage AT1 transports neutral amino acids. Lysine is not a neutral amino acid: it is positively charged.
3. Arginine is a positively charged amino acid. The passage indicates that AT1 transports neutral amino acids, thus not arginine.
4. According to the passage AT1 transports neutral amino acids. AT1 cannot transport glutamate because glutamate is negatively charged.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
1 min 11 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1A
FC1
BCH

Question 8 of 59

Data ID: 831382
Captured: 6/25/2026, 6:24:43 PM

PASSAGE

Passage 2 (Questions 7-11)

Malnutrition is known to increase susceptibility to colitis (colon inflammation). Colitis can also be caused by pellagra (a condition that may result from tryptophan deficiency) and Hartnup disease (a condition resulting from decreased function of the small intestinal neutral amino acid transporter AT1), the latter of which reduces gut absorption of tryptophan. The plasma membrane-embedded domain of the angiotensin I converting enzyme 2 (ACE2) is necessary for AT1 localization to the luminal plasma membrane of small intestinal epithelial cells. AT1 simultaneously transports tryptophan against its concentration gradient and a sodium ion along its concentration gradient into the cytoplasm of intestinal epithelial cells.

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	PFD	22	not determined
	SD + G–T	2	1.4
–/y	SD*	10	0.3
	PFD	20	not determined
	SD + G–T	2	1.0

*Both mice genotypes on SD showed 0% weight loss without DSS.

Adapted from Tatsuo Hashimoto et al., *Nature* ©2012 Macmillan Publishers Limited.

QUESTION + ANSWER CHOICES

Based on the information in the passage, a mature AT1 mRNA is most likely to contain a sequence coding for which genetic factor(s)?

- **A.**
  Signal sequence
- B.
  Introns
- C.
  Promoter
- D.
  Nuclear localization signal

SOLUTION

Solution: The correct answer is **A**.

1. **Normally, the mature RNA sequence for a protein that is secreted or that will locate to the cell membrane will contain the signal sequence. This portion of the mRNA is located in the 5′ region and will signal to the ribosome that translation needs to be continued in the rough endoplasmic reticulum.**
2. Introns are absent in mature mRNA.
3. The promoter is not present in mRNA.
4. The nuclear localization signal is a sequence that tags the protein for it to be transported into the nucleus. Based on the passage, AT1 is a protein transporter located on the cell membrane, not inside the nucleus.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
x
Time Spent:
4 mins 1 sec
Difficulty Level:
Content & Skills:
SIRS2
CC2A
FC2
BIO

Question 9 of 59

Data ID: 831385
Captured: 6/25/2026, 6:24:43 PM

PASSAGE

Passage 2 (Questions 7-11)

Malnutrition is known to increase susceptibility to colitis (colon inflammation). Colitis can also be caused by pellagra (a condition that may result from tryptophan deficiency) and Hartnup disease (a condition resulting from decreased function of the small intestinal neutral amino acid transporter AT1), the latter of which reduces gut absorption of tryptophan. The plasma membrane-embedded domain of the angiotensin I converting enzyme 2 (ACE2) is necessary for AT1 localization to the luminal plasma membrane of small intestinal epithelial cells. AT1 simultaneously transports tryptophan against its concentration gradient and a sodium ion along its concentration gradient into the cytoplasm of intestinal epithelial cells.

To determine how malnutrition promotes colitis, researchers treated wild-type (WT) mice and mice with loss-of-function mutations in the X-linked *Ace2* gene with dextran sodium sulfate (DSS), which increases colitis susceptibility by damaging intestinal epithelial cells. Mice were then fed one of the following three diets:

1. Standard diet (SD);
2. Protein-free diet (PFD); or
3. Gly–Trp (G–T), a dipeptide not requiring AT1 for intestinal uptake.

Mice were then monitored for weight loss as an indicator of colitis (Table 1).

Table 1 The Effects of *Ace2* Genotype and Diet on Colitis and Circulating Tryptophan

Ace2 genotype	Diet	Percent weight loss 4 days after start of DSS treatment (%)	Tryptophan blood levels (relative to control)
+/y (WT)	SD*	2	1.0
	PFD	22	not determined
	SD + G–T	2	1.4
–/y	SD*	10	0.3
	PFD	20	not determined
	SD + G–T	2	1.0

*Both mice genotypes on SD showed 0% weight loss without DSS.

Adapted from Tatsuo Hashimoto et al., *Nature* ©2012 Macmillan Publishers Limited.

QUESTION + ANSWER CHOICES

Which effect did a protein-free diet have on DSS-treated +/y and –/y mice, respectively?

- A.
  Both the +/y and the –/y mice gained weight.
- B.
  The +/y mice lost weight, and the –/y mice gained weight.
- C.
  The +/y mice gained weight, and the –/y mice lost weight.
- **D.**
  Both +/y and the –/y mice lost weight.

SOLUTION

Solution: The correct answer is **D**.

1. Based on Table 1, both mice genotypes lost weight when fed a protein-free diet.
2. Table 1 summarizes the weight loss at four days after the start of the diet and, in both genotypes, there is a loss of weight.
3. Based on Table 1, WT mice lost 22% of their body weight 4 days after the start of the diet, and the mutant mice lost 20%. Thus, both mice lost weight.
4. **Based on Table 1, both mice genotypes lost weight when fed a protein-free diet: WT lost 22% and mutants lost 20%.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 8 secs
Difficulty Level:
Content & Skills:
SIRS4
CC1C
FC1
BIO

Question 10 of 59

Data ID: 831390
Captured: 6/25/2026, 6:24:44 PM

PASSAGE

Passage 2 (Questions 7-11)

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*Both mice genotypes on SD showed 0% weight loss without DSS.

Adapted from Tatsuo Hashimoto et al., *Nature* ©2012 Macmillan Publishers Limited.

QUESTION + ANSWER CHOICES

Pellagra also results from a deficiency of nicotinamide, which is synthesized from tryptophan. Nicotinamide nucleotides are neither oxidized nor reduced during which step of cellular respiration?

- A.
  Glycolysis
- **B.**
  Chemiosmosis
- C.
  Citric acid cycle
- D.
  Electron transport chain

SOLUTION

Solution: The correct answer is **B**.

1. During glycolysis NAD^+ is reduced to form NADH.
2. **Chemiosmosis is the only step of cellular respiration where NAD^+ is neither reduced to form NADH, nor is NADH oxidized to form NAD^+.**
3. During the citric acid cycle, NAD^+ is reduced to form NADH.
4. During the electron transport chain, NADH is oxidized to form NAD^+.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
x
D
Time Spent:
1 min 0 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1D
FC1
BCH

Question 11 of 59

Data ID: 831394
Captured: 6/25/2026, 6:24:44 PM

PASSAGE

Passage 2 (Questions 7-11)

Malnutrition is known to increase susceptibility to colitis (colon inflammation). Colitis can also be caused by pellagra (a condition that may result from tryptophan deficiency) and Hartnup disease (a condition resulting from decreased function of the small intestinal neutral amino acid transporter AT1), the latter of which reduces gut absorption of tryptophan. The plasma membrane-embedded domain of the angiotensin I converting enzyme 2 (ACE2) is necessary for AT1 localization to the luminal plasma membrane of small intestinal epithelial cells. AT1 simultaneously transports tryptophan against its concentration gradient and a sodium ion along its concentration gradient into the cytoplasm of intestinal epithelial cells.

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	SD + G–T	2	1.0

*Both mice genotypes on SD showed 0% weight loss without DSS.

Adapted from Tatsuo Hashimoto et al., *Nature* ©2012 Macmillan Publishers Limited.

QUESTION + ANSWER CHOICES

If a man with a mutant copy of *Ace2* has a child with a woman that is heterozygous for the mutant *Ace2* allele, what is the probability that the child will be a female and homozygous for the mutant *Ace2* allele?

- A.
  0
- **B.**
  0.25
- C.
  0.75
- D.
  1

SOLUTION

Solution: The correct answer is **B**.

1. Based on the pattern of inheritance, a daughter will inherit one of the X chromosomes from her father. If the father carries the mutation on his X chromosome, the daughter will surely carry the mutant chromosome. If the mother is heterozygous, then the daughter has more than 0% chances to be homozygous.
2. **Based on the pattern of inheritance, a daughter will inherit one of the X chromosomes from her father. The daughter will inherit the other X chromosome from her mother. If the mother is heterozygous, the daughter will have a 50% chance of being homozygous. As there is a 50% chance of the offspring being a female, based on the rules of probability, there is 25% of probability (50% × 50%) that the child will be a homozygous mutant daughter.**
3. The scenario describes the conditions for the daughter to have a 25% chance of being homozygous for the mutation. There is no scenario possible for the daughter of having a 75% chance of being homozygous for the mutation.
4. For the daughter to have a 100% chance of being a homozygous mutant, the father must carry the mutation and the mother must be homozygous for the mutation. In this case, however, the mother is heterozygous.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
2 mins 2 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1C
FC1
BIO

Question 12 of 59

Data ID: 831401
Captured: 6/25/2026, 6:24:45 PM

QUESTION + ANSWER CHOICES

During the initiation of muscle contraction, mysosin binds actin after troponin binds to which ion?

- A.
  H^+
- B.
  K^+
- C.
  Na^+
- **D.**
  Ca^{2+}

SOLUTION

Solution: The correct answer is **D**.

1. H^+ does not play a role in myosin binding to actin during muscle contraction.
2. K^+ ions play a role in the maintenance of membrane potential, not the binding of myosin to actin, during muscle contraction.
3. Na^+ ions are involved in the maintenance of the membrane potential, not in the binding of myosin to actin.
4. **Ca^{2+} ions play a critical role in myosin-actin binding during skeletal muscle contraction. Ca^{2+} binds to troponin and allows tropomyosin to move, freeing the site of interaction between actin and myosin.**

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✔
Time Spent:
44 secs
Difficulty Level:
Content & Skills:
SIRS1
CC3B
FC3
BIO

Question 13 of 59

Data ID: 831405
Captured: 6/25/2026, 6:24:46 PM

QUESTION + ANSWER CHOICES

What is the total number of fused rings present in a steroid?

- A.
  1
- B.
  2
- **C.**
  4
- D.
  6

SOLUTION

Solution: The correct answer is **C**.

1. A single ring can be found in a monoterpene, not a steroid.
2. An indole has two fused rings, but it is not a steroid.
3. **A steroid is composed of a 6-6-6-5 fused ring assembly, which is a total of 4 fused rings.**
4. Some natural products have 6 fused rings, but they are not steroids.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
11 secs
Difficulty Level:
Content & Skills:
SIRS1
CC2A
FC2
OCH

Question 14 of 59

Data ID: 831411
Captured: 6/25/2026, 6:24:46 PM

QUESTION + ANSWER CHOICES

Epilepsy may result in motor seizures due to massive synchronous firing of neurons in a small area of the cerebral cortex (the *epileptic focus*). Excitation spreads from the focus, involving an increasingly larger area of the cortex. A drug for the treatment of epilepsy would be most effective if it caused which of the following changes in the epileptic focus?

- **A.**
  An increase in the neuron-firing threshold
- B.
  An increase in extracellular Na^+ concentration
- ~~C.~~
  ~~A decrease in axon-membrane permeability to negative ions~~
- D.
  ~~A decrease in the length of the depolarization stage~~

SOLUTION

Solution: The correct answer is **A**.

1. **An increase in the threshold would make it more challenging for a neuron to fire an action potential. This will reduce the frequency of downstream neuron depolarizations, resulting in relief of the symptoms.**
2. An increase in the concentration of Na^+ outside the cell will not help reduce the symptoms, because there will still be a flux of ions inside the cell when the firing starts, causing continuous depolarization.
3. Anion influx hyperpolarizes a neuron, reducing likelihood of action potential firing. Therefore, a decrease in axonal-membrane permeability to negative ions will not provide a relief as it will increase action potential firing.
4. A decrease in length of the depolarization stage will allow for a higher frequency of action potential firing. This would worsen the situation instead of improving it.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
56 secs
Difficulty Level:
Content & Skills:
SIRS2
CC3A
FC3
BIO

Question 15 of 59

Data ID: 831414
Captured: 6/25/2026, 6:24:47 PM

QUESTION + ANSWER CHOICES

When concentrated urine is being produced, in which of the following regions of the kidney will the glomerular filtrate reach its highest concentration?

- A.
  Proximal convoluted tubule
- B.
  Distal convoluted tubule
- C.
  Cortical portion of the collecting duct
- **D.**
  Medullary portion of the collecting duct

SOLUTION

Solution: The correct answer is **D**.

1. The proxymal convoluted tubule is the first tubule where the glomerular filtrate passes through. About 65% of the glomerular filtrate is reabsorbed in the proximal collecting tubule. Thus, this is not the region where the glomerular filtrate is the most concentrated.
2. The distal tubule contains relatively dilute glomerular filtrate. Water reabsorption can take place in the collecting duct, which concentrates the filtrate.
3. The collecting duct is the last portion of the tubules where reabsorption of water and salts can occur. The glomerular filtrate is, however, slightly less concentrated in the cortical portion of the collecting duct than in the medullary portion.
4. **The collecting duct is the final structure in which water reabsorption occurs, which concentrates filtrate. The medullary portion of the collecting duct is the last portion of the tubules where reabsorption can occur. In the portion of the tubule that follows, there will be no more reabsorption. Thus, the medullary portion of the collecting duct contains the most concentrated glomerular filtrate that will correspond to the urine.**

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
x
C
D
Time Spent:
33 secs
Difficulty Level:
Content & Skills:
SIRS1
CC3B
FC3
BIO

Question 16 of 59

Data ID: 831420
Captured: 6/25/2026, 6:24:48 PM

PASSAGE

Passage 3 (Questions 16-19)

The cellular membrane potential is primarily determined by the concentrations of Na^+ and K^+. The Na^+K^+ ATPase is a membrane protein that maintains the concentrations of these ions within the cell. In addition, the concentration gradient established by the Na^+K^+ ATPase is used as the driving force in the cotransport of other biological molecules. Recent studies have shown that there is a correlation between decreased heart function and decreased Na^+K^+ ATPase activity.

An important question regarding membrane proteins is how the composition of the membrane affects the protein function. Two major factors that are modulated by membrane composition are fluidity and thickness. Membrane composition is changed by both the structure of the component phospholipids and the amount of embedded cholesterol. For instance, equations 1 and 2 can be used to calculate membrane thickness *d* in angstroms when the membrane is composed of saturated lipids (Equation 1) or monounsaturated lipids (Equation 2), where n_c represents the number of carbon atoms in the alkyl chains.

$$d_{sat} = 1.75(n_c - 1)$$

Equation 1

$$d_{unsat} = 1.75(n_c - 2.6)$$

Equation 2

In order to study how membrane composition affects protein function, Na^+K^+ ATPase was reconstituted into artificially engineered lipid bilayers called liposomes. The liposomes used in this study were composed of phosphatidylcholine (PC). Two experiments were conducted.

Experiment 1

Na^+K^+ ATPase was reconstituted with either monounsaturated PC whose acyl chain length is 14 (14:1) or saturated PC of the same chain length (14:0). The temperature dependence of Na^+K^+ ATPase activity was measured in the presence and absence of 40 mol% cholesterol (Figure 1).

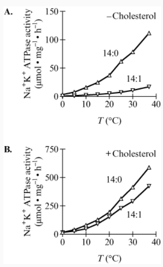

Figure 1	Temperature dependence of reconstituted Na^+K^+ ATPase activity in the absence (A) or presence (B) of cholesterol

Experiment 2

Na^+K^+ ATPase was reconstituted with monounsaturated PC of various acyl chain lengths (n_c), and the Na^+K^+ ATPase activity was measured for each in the presence and absence of 40 mol% cholesterol at 25°C (Figure 2).

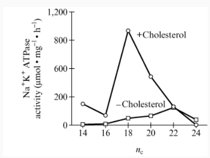

Figure 2	Acyl chain length dependence of Na^+K^+ ATPase activity

QUESTION + ANSWER CHOICES

What aspect of Experiment 1 does NOT address whether membrane composition has an effect on Na^+K^+ ATPase activity? The activity of the Na^+K^+ ATPase:

- A.
  showed less temperature dependence in the 14:1 liposome than the 14:0 liposome.
- B.
  was highest in the 14:0 liposomes at all temperatures.
- **C.**
  increased with temperature in both the 14:1 liposome and the 14:0 liposome.
- D.
  was greater at all temperatures when cholesterol was present.

SOLUTION

Solution: The correct answer is **C**.

1. Figure 1 shows that Na^+K^+ ATPase activity in 14:1 liposome is indeed less dependent on temperature than in the 14:0 liposome.
2. Data obtained in Experiment 1 show that Na^+K^+ ATPase activity in the 14:0 liposome is indeed highest at all temperatures.
3. **In kinetic studies, protein activity is expected to increase with temperature, assuming the protein does not thermally denature. Therefore, the researchers did not raise the temperature to study the effects of temperature on protein activity, they instead raised the temperature to study how membrane fluidity plays a role in protein activity.**
4. In Experiment 1, researchers analyzed the temperature dependence of Na^+K^+ ATPase activity in the presence and absence of cholesterol and showed that that it was greater at all temperatures when cholesterol was present.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
x
Time Spent:
4 mins 56 secs
Difficulty Level:
Content & Skills:
SIRS3
CC1A
FC1
BCH

Question 17 of 59

Data ID: 831423
Captured: 6/25/2026, 6:24:48 PM

PASSAGE

Passage 3 (Questions 16-19)

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An important question regarding membrane proteins is how the composition of the membrane affects the protein function. Two major factors that are modulated by membrane composition are fluidity and thickness. Membrane composition is changed by both the structure of the component phospholipids and the amount of embedded cholesterol. For instance, equations 1 and 2 can be used to calculate membrane thickness *d* in angstroms when the membrane is composed of saturated lipids (Equation 1) or monounsaturated lipids (Equation 2), where n_c represents the number of carbon atoms in the alkyl chains.

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Equation 1

$$d_{unsat} = 1.75(n_c - 2.6)$$
Equation 2

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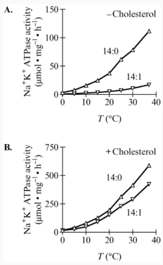

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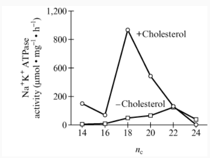

Figure 2	Acyl chain length dependence of Na^+K^+ ATPase activity

QUESTION + ANSWER CHOICES

Which approach does NOT measure the activity of the Na^+K^+ ATPase?

- A.
  Measuring the rate of ATP hydrolysis
- **B.**
  Measuring the free energy of the ion transport
- C.
  Measuring the rate of ADP production
- D.
  Measuring the change in ion concentration within the liposome

SOLUTION

Solution: The correct answer is **B**.

1. The activity of the Na^+K^+ ATPase is directly related to the rate of ATP hydrolysis.
2. **Free energy does not correlate with enzyme activity because it is NOT a measure of reaction rate/activity.**
3. The rate of ADP production does correlate with ATPase activity because ADP is a product of Na^+K^+ ATPase catalysis.
4. Na^+K^+ ATPase catalyzes ion transport through the liposome, therefore a change in ion concentration correlates with Na^+K^+ ATPase activity.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
31 secs
Difficulty Level:
Content & Skills:
SIRS3
CC1D
FC1
GCH

Question 18 of 59

Data ID: 831426
Captured: 6/25/2026, 6:24:49 PM

PASSAGE

Passage 3 (Questions 16-19)

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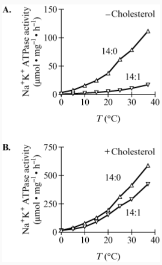

Figure 1	Temperature dependence of reconstituted Na^+K^+ ATPase activity in the absence (A) or presence (B) of cholesterol

Experiment 2

Na^+K^+ ATPase was reconstituted with monounsaturated PC of various acyl chain lengths (n_c), and the Na^+K^+ ATPase activity was measured for each in the presence and absence of 40 mol% cholesterol at 25°C (Figure 2).

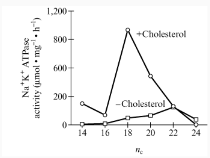

Figure 2	Acyl chain length dependence of Na^+K^+ ATPase activity

QUESTION + ANSWER CHOICES

What is the function of the Na^+K^+ ATPase during a neuronal action potential?

- A.
  Stimulation of the action potential
- B.
  Depolarization of the membrane
- C.
  Hyperpolarization of the membrane
- **D.**
  Restoration of the resting potential

SOLUTION

Solution: The correct answer is **D**.

1. The action potential is stimulated by the release of a neurotransmitter that binds to ligand gated channels, not the Na^+/K^+ ATPase.
2. Membrane depolarization is due to the opening of Na^+ channels, a process that is independent from the Na^+/K^+ ATPase.
3. Hyperpolarization of the cell membrane occurs when K^+ channels open and Na^+ channels close. This process is independent of the Na^+/K^+ ATPase.
4. **The restoration and maintenance of the resting potential relies on Na^+/K^+ ATPase. This is achieved by moving three Na^+ out of the cell for every two K^+ ions that are brought into the cell.**

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
x
C
D
Time Spent:
32 secs
Difficulty Level:
Content & Skills:
SIRS1
CC3A
FC3
BIO

Question 19 of 59

Data ID: 831429
Captured: 6/25/2026, 6:24:49 PM

PASSAGE

Passage 3 (Questions 16-19)

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Equation 1

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Equation 2

In order to study how membrane composition affects protein function, Na^+K^+ ATPase was reconstituted into artificially engineered lipid bilayers called liposomes. The liposomes used in this study were composed of phosphatidylcholine (PC). Two experiments were conducted.

Experiment 1

Na^+K^+ ATPase was reconstituted with either monounsaturated PC whose acyl chain length is 14 (14:1) or saturated PC of the same chain length (14:0). The temperature dependence of Na^+K^+ ATPase activity was measured in the presence and absence of 40 mol% cholesterol (Figure 1).

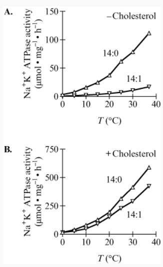

Figure 1	Temperature dependence of reconstituted Na^+K^+ ATPase activity in the absence (A) or presence (B) of cholesterol

Experiment 2

Na^+K^+ ATPase was reconstituted with monounsaturated PC of various acyl chain lengths (n_c), and the Na^+K^+ ATPase activity was measured for each in the presence and absence of 40 mol% cholesterol at 25°C (Figure 2).

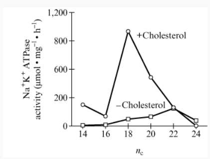

Figure 2	Acyl chain length dependence of Na^+K^+ ATPase activity

QUESTION + ANSWER CHOICES

In which direction does the Na^+K^+ ATPase transport ions across the cell membrane upon ATP hydrolysis?

- **A.**
  Na^+ is transported out of the cell; K^+ is transported into the cell.
- B.
  Na^+ is transported into the cell; K^+ is transported out of the cell.
- C.
  Both Na^+ and K^+ are transported into the cell.
- D.
  Both Na^+ and K^+ are transported out of the cell.

SOLUTION

Solution: The correct answer is **A**.

1. **Upon ATP hydrolysis, three Na^+ are transported outside the cell and two K^+ are transported inside the cell against their concentration gradient.**
2. Na^+ and K^+ are transported against their concentration gradient, thus Na^+ is transported outside the cell and K^+ is transported inside the cell, not the opposite.
3. Na^+ and K^+ are transported against their concentration gradient, thus Na^+ is transported outside the cell and K^+ is transported inside the cell.
4. If both Na^+ and K^+ were transported out of the cell, K^+ would not be transported against its concentration gradient. Instead, both Na^+ and K^+ are transported against their concentration gradient. Thus Na^+ is transported outside the cell and K^+ is transported inside the cell.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
39 secs
Difficulty Level:
Content & Skills:
SIRS1
CC2A
FC2
BIO

Question 20 of 59

Data ID: 831435
Captured: 6/25/2026, 6:24:50 PM

PASSAGE

Passage 4 (Questions 20-23)

Diffuse large B-cell lymphoma (DLBCL) is the most common subtype of malignant lymphoma. The most common feature of the molecular pathogenesis of DLBCL is the constant activation of the NF-κB pathway. The NF-κB proteins p65 and cRel are transcription factors that can be continuously activated when mutations occur to proteins upstream in the signaling pathway (Figure 1). Their activation leads to increased cell proliferation and suppression of apoptosis. Activation occurs when the inhibitory IκBα protein is degraded. This frees the nuclear localization sequence (NLS) on p65. The NLS is composed of four amino acids (KRKR) and interacts with an importin on the nuclear envelope.

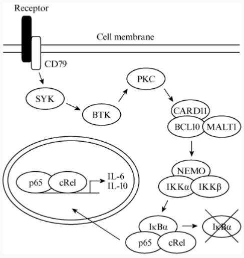

Figure 1	The major proteins in the NF-κB signaling pathway (Note: The IκBα is crossed-out to show it is degraded in response to the upstream signal.)

Drugs that target the variant signaling proteins may represent an effective therapeutic tool in the treatment of DLBCL. One such drug is a selective protein kinase C (PKC) inhibitor called Sotrastaurin (STN, Figure 2).

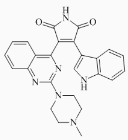

Figure 2	The structure of STN

STN was tested in four different DLBCL cell lines. Each cell line had a different mutation in the NF-κB pathway that resulted in the overproduction of interleukins 6 and 10 (IL-6 and IL-10, respectively). The cell lines HBL1, TMD8, and Ly10 all had mutations in the CD79 protein, and the cell line Ly3 had a mutation in the CARD11 protein. These mutations resulted in continuous activation of these proteins regardless of upstream signals. The effectiveness of STN was determined by the concentration dependent drop in IL-6 mRNA levels (Figure 3). It was also found that STN treatment caused TMD8 and HBL1 cells to undergo cell-cycle arrest that was characterized by a decrease in the number of cells with 4N DNA content.

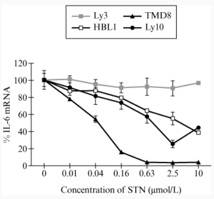

Figure 3	The STN concentration dependent expression of IL-6 mRNA in the DLBCL cell lines discussed in the passage

QUESTION + ANSWER CHOICES

Upon activation, p65 and cRel control the level of IL-6 mRNA by:

- A.
  binding RNA.
- **B.**
  binding DNA.
- C.
  replicating RNA.
- D.
  replicating DNA.

SOLUTION

Solution: The correct answer is **B**.

1. Transcription factors regulate the expression of other genes by binding to DNA, not RNA.
2. **p65 and cRel are transcription factors and regulate the expression of other genes by binding to the promoter or the enhancer of the gene located on the DNA.**
3. It is DNA, and not RNA, that interacts with the transcription factors.
4. Transcription factors regulate the expression of specific genes, they do not replicate the entire genome.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
x
D
Time Spent:
3 mins 27 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1B
FC1
BIO

Question 21 of 59

Data ID: 831438
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PASSAGE

Passage 4 (Questions 20-23)

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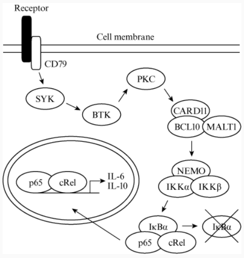

Figure 1	The major proteins in the NF-κB signaling pathway (Note: The IκBα is crossed-out to show it is degraded in response to the upstream signal.)

Drugs that target the variant signaling proteins may represent an effective therapeutic tool in the treatment of DLBCL. One such drug is a selective protein kinase C (PKC) inhibitor called Sotrastaurin (STN, Figure 2).

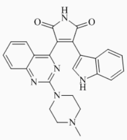

Figure 2	The structure of STN

STN was tested in four different DLBCL cell lines. Each cell line had a different mutation in the NF-κB pathway that resulted in the overproduction of interleukins 6 and 10 (IL-6 and IL-10, respectively). The cell lines HBL1, TMD8, and Ly10 all had mutations in the CD79 protein, and the cell line Ly3 had a mutation in the CARD11 protein. These mutations resulted in continuous activation of these proteins regardless of upstream signals. The effectiveness of STN was determined by the concentration dependent drop in IL-6 mRNA levels (Figure 3). It was also found that STN treatment caused TMD8 and HBL1 cells to undergo cell-cycle arrest that was characterized by a decrease in the number of cells with 4N DNA content.

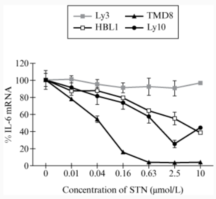

Figure 3	The STN concentration dependent expression of IL-6 mRNA in the DLBCL cell lines discussed in the passage

QUESTION + ANSWER CHOICES

Considering the structure of STN, what is the most likely mechanism for its entry into the cell?

- A.
  Active transport
- B.
  Receptor mediated endocytosis
- **C.**
  Diffusion directly through the membrane
- D.
  Passage through an ion channel

SOLUTION

Solution: The correct answer is **C**.

1. The structure of STN shows that it is a hydrophobic molecule. These type of molecules can pass through the membrane by simple diffusion. In addition the transport is not carried against a concentration gradient which would require an active transport.
2. The structure of STN shows that it is a hydrophobic molecule. These types of molecules can pass through the membrane by simple diffusion and therefore do not need a transporter.
3. **The structure of STN shows that it is a hydrophobic molecule. These types of molecules pass through the membrane by simple diffusion.**
4. The structure of STN shows that it is a hydrophobic molecule, not an ion.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
2 mins 6 secs
Difficulty Level:
Content & Skills:
SIRS2
CC2A
FC2
BCH

Question 22 of 59

Data ID: 831441
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PASSAGE

Passage 4 (Questions 20-23)

Diffuse large B-cell lymphoma (DLBCL) is the most common subtype of malignant lymphoma. The most common feature of the molecular pathogenesis of DLBCL is the constant activation of the NF-κB pathway. The NF-κB proteins p65 and cRel are transcription factors that can be continuously activated when mutations occur to proteins upstream in the signaling pathway (Figure 1). Their activation leads to increased cell proliferation and suppression of apoptosis. Activation occurs when the inhibitory IκBα protein is degraded. This frees the nuclear localization sequence (NLS) on p65. The NLS is composed of four amino acids (KRKR) and interacts with an importin on the nuclear envelope.

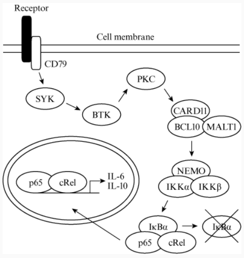
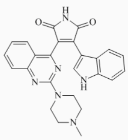


Figure 1	The major proteins in the NF-κB signaling pathway (Note: The IκBα is crossed-out to show it is degraded in response to the upstream signal.)

Drugs that target the variant signaling proteins may represent an effective therapeutic tool in the treatment of DLBCL. One such drug is a selective protein kinase C (PKC) inhibitor called Sotrastaurin (STN, Figure 2).



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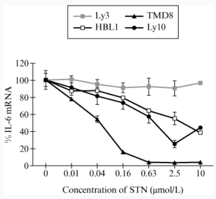

Figure 3	The STN concentration dependent expression of IL-6 mRNA in the DLBCL cell lines discussed in the passage

QUESTION + ANSWER CHOICES

Assuming no mutations to the signaling pathway described in the passage, what event directly activates CARD11?

- A.
  Proteolytic cleavage
- B.
  GTP binding
- C.
  Membrane association
- **D.**
  Phosphorylation

SOLUTION

Solution: The correct answer is **D**.

1. The passage implies that phosphorylation by PKC and not proteolytic cleavage activates CARD11.
2. The passage implies that phosphorylation, and not GTP binding, activates CARD11.
3. The passage implies that phosphorylation, and not membrane association, activates CARD11.
4. **Based on the passage, the protein that activates CARD11 is a protein kinase. This implies that phosphorylation is the activating event.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 9 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1A
FC1
BCH

Question 23 of 59

Data ID: 831442
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PASSAGE

Passage 4 (Questions 20-23)

Diffuse large B-cell lymphoma (DLBCL) is the most common subtype of malignant lymphoma. The most common feature of the molecular pathogenesis of DLBCL is the constant activation of the NF-κB pathway. The NF-κB proteins p65 and cRel are transcription factors that can be continuously activated when mutations occur to proteins upstream in the signaling pathway (Figure 1). Their activation leads to increased cell proliferation and suppression of apoptosis. Activation occurs when the inhibitory IκBα protein is degraded. This frees the nuclear localization sequence (NLS) on p65. The NLS is composed of four amino acids (KRKR) and interacts with an importin on the nuclear envelope.

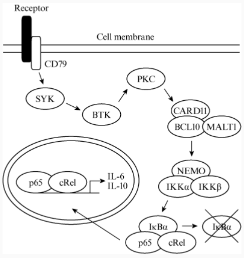

Figure 1	The major proteins in the NF-κB signaling pathway (Note: The IκBα is crossed-out to show it is degraded in response to the upstream signal.)

Drugs that target the variant signaling proteins may represent an effective therapeutic tool in the treatment of DLBCL. One such drug is a selective protein kinase C (PKC) inhibitor called Sotrastaurin (STN, Figure 2).

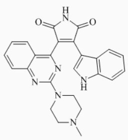

Figure 2	The structure of STN

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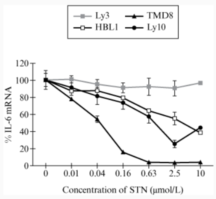

Figure 3	The STN concentration dependent expression of IL-6 mRNA in the DLBCL cell lines discussed in the passage

QUESTION + ANSWER CHOICES

According to the passage, what is most likely correct about the genes that are expressed as a result of the constant activation of the NF-κB pathway?

- A.
  They cause disruption of the mitochondria.
- **B.**
  They contain a p65/cRel binding site in their promoter region.
- C.
  They have accumulated mutations that alter function.
- D.
  They bind to STN in the cytoplasm.

SOLUTION

Solution: The correct answer is **B**.

1. The passage does not describe the role of NF-κB in mitochondria.
2. **Based on the passage, activation of the NF-κB pathway results in translocation of the transcription factor subunits p65/cRel in the nucleus, where they regulate the expression of downstream target genes. Expression regulation is possible because the downstream target genes contain in their promoter or enhancer, DNA regions that are able to interact with p65 and cRel.**
3. Genes downstream of NF-κB signaling have not accumulated mutations; these genes have only been activated.
4. STN is a protein kinase inhibitor, and it inhibits IL-6 expression by acting upstream of the NF-κB pathway: there is no binding to STN in the cytoplasm.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
<u>x</u>
B
C
D
Time Spent:
40 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1B
FC1
BIO

Question 24 of 59

Data ID: 831451
Captured: 6/25/2026, 6:24:52 PM

PASSAGE

Passage 5 (Questions 24-27)

P-glycoprotein (P-gp) is a plasma membrane ABC transporter that has broad substrate specificity for hydrophobic compounds. P-gp is responsible for multidrug resistance (MDR) in tumor cells by preventing drugs from accumulating to cytotoxic levels. Studies show that P-gp overproduction in the membrane of tumor cells produces an MDR phenotype.

It has been shown that P-gp exhibits ATPase activity and is localized in cholesterol-rich domains within the membrane. To determine whether a functional link exists between P-gp and cholesterol, researchers conducted two experiments.

Experiment 1

Methyl-β-cyclodextrin (MBCD) is a cyclic oligosaccharide that binds and removes cholesterol from cell membranes. MBCD was added to cells that either overexpressed P-gp (MDR) or were P-gp free (control). The ATPase activity was measured at 37°C, and the reactions were supplemented with 1 mM ATP (Figure 1).

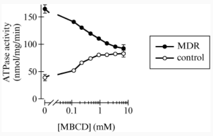

Figure 1	The MBCD-dependent ATPase activity of MDR and control cells

Experiment 2

The ATPase activity of P-gp was measured in the presence of increasing concentrations of cholesterol in MDR cells that were never treated with MBCD (native MDR) and MDR cells that were treated with MBCD for 1 hour at 37°C.

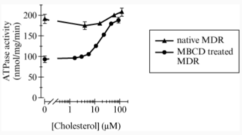

Figure 2	The cholesterol-dependent ATPase activity of native MDR and MDR cells pre-treated with MBCD for 1 hour at 37°C

Adapted from A. Garrigues et al., *PNAS* ©2002 the Authors.

QUESTION + ANSWER CHOICES

Which mechanism best describes how P-gp facilitates drug resistance?

- A.
  P-gp binds to antitumor drugs in the presence of ATP and degrades the drugs.
- **B.**
  P-gp serves as a pump and uses active transport to move antitumor drugs outside the cell.
- C.
  P-gp prevents the entry of anti-tumor drugs into the cell.
- D.
  P-gp causes increased membrane permeability, which causes antitumor drugs to exit the cell.

SOLUTION

Solution: The correct answer is **B**.

1. The passage notes that P-gp is an ABC transporter protein, not a binding protein.
2. **The passage notes that P-gp is an ABC transporter protein with ATPase activity. These transporters use ATP to actively transport molecules such as drugs out of the cell.**
3. The passage notes that P-gp is an ABC transporter protein. Transporter proteins facilitate, rather than prevent, transfer of molecules across the membrane bilayer.
4. The passage notes that P-gp is an ABC transporter protein. Transporter proteins are not involved in increased membrane permeability, they function to carry molecules across membranes.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 0 secs
Difficulty Level:
Content & Skills:
SIRS2
CC2A
FC2
BCH

Question 25 of 59

Data ID: 831454
Captured: 6/25/2026, 6:24:53 PM

PASSAGE

Passage 5 (Questions 24-27)

P-glycoprotein (P-gp) is a plasma membrane ABC transporter that has broad substrate specificity for hydrophobic compounds. P-gp is responsible for multidrug resistance (MDR) in tumor cells by preventing drugs from accumulating to cytotoxic levels. Studies show that P-gp overproduction in the membrane of tumor cells produces an MDR phenotype.

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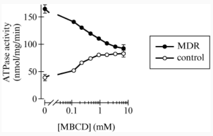

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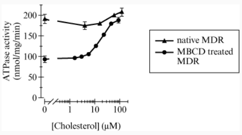

Figure 2	The cholesterol-dependent ATPase activity of native MDR and MDR cells pre-treated with MBCD for 1 hour at 37°C

QUESTION + ANSWER CHOICES

The formation of which reaction products is increased in MBCD-treated MDR cells after exposure to 100 μM cholesterol?

- A.
  AMP + ADP
- **B.**
  ADP + P_i
- C.
  ATP + P_i
- D.
  ADP + ATP

SOLUTION

Solution: The correct answer is **B**.

1. Data in Figure 2 show that exposure of cholesterol-depleted MDR cells to 100 μM cholesterol results in increased ATPase activity (ATP hydrolysis). ATP hydrolysis produces ADP and inorganic phosphate, not AMP.
2. **Figure 2 shows that exposure of cholesterol-depleted MDR cells to 100 μM cholesterol results in complete restoration of ATPase activity (ATP hydrolysis). ATP hydrolysis produces ADP and inorganic phosphate.**
3. Data in Figure 2 show that exposure of cholesterol-depleted MDR cells to 100 μM cholesterol results in increased ATP hydrolysis. ATP is the reagent in the hydrolysis, not a product.
4. Exposure of cholesterol-depleted MDR cells to 100 μM cholesterol results in increased ATP hydrolysis. While ADP is a product of the hydrolysis, ATP is the reagent in the hydrolysis, not a product.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
2 mins 49 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1D
FC1
BCH

Question 26 of 59

Data ID: 831457
Captured: 6/25/2026, 6:24:54 PM

PASSAGE

Passage 5 (Questions 24-27)

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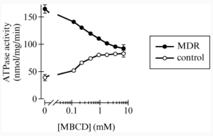

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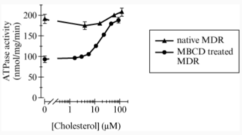


Figure 2	The cholesterol-dependent ATPase activity of native MDR and MDR cells pre-treated with MBCD for 1 hour at 37°C

QUESTION + ANSWER CHOICES

What is the most likely location of P-gp within the plasma membrane?

- ~~A.~~
  ~~Associated with lipids on the cytoplasmic side only~~
- ~~B.~~
  ~~Associated with lipids on the extracellular side only~~
- C.
  Peripheral to the plasma membrane
- **D.**
  Within a lipid raft

SOLUTION

Solution: The correct answer is **D**.

1. Based on the passage, P-gp is localized in cholesterol-rich domains within the membrane which are known as lipid rafts. These structures are not associated with lipids on the cytoplasmic side of the membrane, but are part of the plasma membrane.
2. Based on the passage, P-gp is localized in cholesterol-rich domains within the membrane which are known as lipid rafts. These structures are not associated with lipids on the extracellular side of the membrane, but are part of the plasma membrane.
3. The passage specifically states that P-gp is localized in cholesterol-rich domains within the membrane, not peripheral to the plasma membrane. Lipid rafts are cholesterol-rich domains within the membrane.
4. **The passage states the P-gp is found in cholesterol rich domains, pointing to lipid rafts.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
1 min 18 secs
Difficulty Level:
Content & Skills:
SIRS1
CC2A
FC2
BCH

Question 27 of 59

Data ID: 831459
Captured: 6/25/2026, 6:24:54 PM

PASSAGE

Passage 5 (Questions 24-27)

P-glycoprotein (P-gp) is a plasma membrane ABC transporter that has broad substrate specificity for hydrophobic compounds. P-gp is responsible for multidrug resistance (MDR) in tumor cells by preventing drugs from accumulating to cytotoxic levels. Studies show that P-gp overproduction in the membrane of tumor cells produces an MDR phenotype.

It has been shown that P-gp exhibits ATPase activity and is localized in cholesterol-rich domains within the membrane. To determine whether a functional link exists between P-gp and cholesterol, researchers conducted two experiments.

Experiment 1

Methyl-β-cyclodextrin (MBCD) is a cyclic oligosaccharide that binds and removes cholesterol from cell membranes. MBCD was added to cells that either overexpressed P-gp (MDR) or were P-gp free (control). The ATPase activity was measured at 37°C, and the reactions were supplemented with 1 mM ATP (Figure 1).

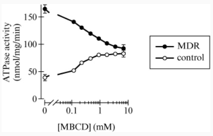


Figure 1	The MBCD-dependent ATPase activity of MDR and control cells

Experiment 2

The ATPase activity of P-gp was measured in the presence of increasing concentrations of cholesterol in MDR cells that were never treated with MBCD (native MDR) and MDR cells that were treated with MBCD for 1 hour at 37°C.

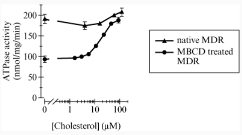


Figure 2	The cholesterol-dependent ATPase activity of native MDR and MDR cells pre-treated with MBCD for 1 hour at 37°C

QUESTION + ANSWER CHOICES

MBCD is composed of which type of molecule?

- A.
  Amino acids
- B.
  Fatty acids
- C.
  Nucleotides
- **D.**
  Carbohydrates

SOLUTION

Solution: The correct answer is **D**.

1. Oligosaccharides are composed of carbohydrates, not amino acids.
2. Oligosaccharides are composed of carbohydrates, not fatty acids.
3. Oligosaccharides are composed of carbohydrates, not nucleotides.
4. **The passage notes that MBCD is a cyclic oligosaccharide, indicating that it is composed of carbohydrate molecules.**

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✔
Time Spent:
1 min 20 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1D
FC1
BCH

Question 28 of 59

Data ID: 831464
Captured: 6/25/2026, 6:24:55 PM

QUESTION + ANSWER CHOICES

Colchicine is a drug that prevents the formation of microtubules. Which of the following mitotic processes would NOT occur after exposure to this drug?

- A.
  Formation of new cell plate
- B.
  Replication of DNA during interphase
- C.
  Breakdown of the nucleolus during prophase
- **D.**
  Movement of the chromosomes toward opposite poles of the cell during anaphase

SOLUTION

Solution: The correct answer is **D**.

1. Microtubules do not play a role in the formation of the cell plate.
2. DNA replication is not regulated by microtubules.
3. Nucleolus breakdown during prophase is not regulated by microtubules.
4. **Microtubules bind to chromosomes at the level of the kinetochore and regulate their migration toward the opposite poles of the cell during anaphase.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
18 secs
Difficulty Level:
Content & Skills:
SIRS2
CC2C
FC2
BIO

Question 29 of 59

Data ID: 831469
Captured: 6/25/2026, 6:24:55 PM

QUESTION + ANSWER CHOICES

The initial filtration step in the glomerulus of the mammalian kidney occurs primarily by:

- **A.**
  passive flow due to a pressure difference.
- B.
  passive flow resulting from a countercurrent exchange system.
- C.
  active transport of water, followed by movement of electrolytes along a resulting concentration gradient.
- D.
  active transport of electrolytes, followed by passive flow of water along the resulting osmolarity gradient.

SOLUTION

Solution: The correct answer is **A**.

1. **There are three pressures that work together to regulate filtration in the glomerulus: glomerular capillary pressure, capsular hydrostatic pressure, and blood colloid osmotic pressure. The glomerular capillary pressure will force filtrate from a capillary into Bowman's capsule; the other two forces promote movement of the filtrate in the opposite direction.**
2. It is not a countercurrent exchange system; it is the difference in pressures that regulates the formation of the filtrate.
3. Water is not actively transported from the glomerulus into Bowman's capsule.
4. Electrolytes are not actively transported from the glomerulus into Bowman's capsule.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

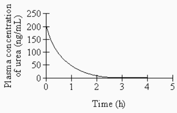
RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
x
Time Spent:
54 secs
Difficulty Level:
Content & Skills:
SIRS1
CC3B
FC3
BIO

Question 30 of 59

Data ID: 831472
Captured: 6/25/2026, 6:24:56 PM

QUESTION + ANSWER CHOICES



The half-life of urea is the time it takes for its concentration to fall by one-half. The graph illustrates the clearance of urea from an initial plasma concentration of 200 ng/mL. If it takes five half-lives for urea to be considered eliminated from the body, how long did it take to eliminate the urea?

- A.
  1.5 h
- **B.**
  2.5 h
- C.
  3.5 h
- D.
  4.0 h

SOLUTION

Solution: The correct answer is **B**.

1. Based on the graph, in 1.5 h, the concentration of urea is down to 25 ng/mL which corresponds to 3 half-lives, not 5.
2. **Based on the graph, it takes 0.5 h for urea concentration to be reduced by half, thus for 5 half-lives, 2.5 hours are required ($0.5h \times 5 = 2.5h$).**
3. The time of 3.5 h corresponds to 7 half-lives, not 5.
4. The time of 4.0 h corresponds to 8 half-lives not 5.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
57 secs
Difficulty Level:
Content & Skills:
SIRS4
CC3B
FC3
BIO

Question 31 of 59

Data ID: 831475
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QUESTION + ANSWER CHOICES

Enzymes alter the rate of chemical reactions by all of the following methods EXCEPT:

- A.
  co-localizing substrates.
- ~~B.~~
  ~~altering local pH.~~
- C.
  ~~altering substrate shape.~~
- **D.**
  altering substrate primary structure.

SOLUTION

Solution: The correct answer is **D**.

1. The enzyme needs to co-localize with the substrates to be able to modify the rate of chemical reactions.
2. It is one of the enzyme functions to change the pH of the environment so that the reaction can occur.
3. The enzyme will alter the shape of the substrate to modify the rate of chemical reactions.
4. **Altering the primary structure of the substrate is not a way for the enzyme to modify the rate of chemical reaction.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
1 min 12 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1A
FC1
BIO

Question 32 of 59

Data ID: 831478
Captured: 6/25/2026, 6:24:57 PM

PASSAGE

Passage 6 (Questions 32-35)

The amino acids from which most proteins are made are called the common amino acids. Most share the general formula shown in Figure 1.

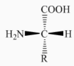

Figure 1

Amino acids can be classified on the basis of the chemical properties of their R groups. Some have hydrophobic, nonpolar R groups (e.g., valine: R = $-CH(CH_3)_2$); some have polar, uncharged R groups and are hydrophilic (e.g., serine: R = $-CH_2OH$); and others have polar, hydrophilic R groups that are acidic (e.g., aspartic acid: R = $-CH_2COOH$) or basic (e.g., lysine: R = $-CH_2CH_2CH_2CH_2NH_2$).

Ornithine (R = $-CH_2CH_2CH_2NH_2$) is an amino acid that is found in cells, but not incorporated into proteins. In mammals and many other organisms, it can be converted to diaminobutane via the ornithine decarboxylase reaction, an early event in cell division. (See Equation 1.)

$$\text{ornithine} + H_2O \rightarrow H_2NCH_2CH_2CH_2CH_2NH_2 + HCO_3^-$$

Equation 1

In the laboratory, the rate of this reaction is measured by an enzyme assay that uses ornithine in which the carboxylic acid carbon is radioactively labeled. The reaction is terminated by acidification of the reaction mixture, which converts the HCO_3^- ion to CO_2 and H_2O.

Thin-layer chromatography is an effective way of separating amino acids. Table 1 below shows R_f values for four amino acids using various solvents. "Orn" represents ornithine. "X," "Y," and "Z" represent three other amino acids.

Table 1

Amino acid	Solvent 1	Solvent 2	Solvent 3
Orn	0.04	0.14	0.24
X	0.15	0.47	0.74
Y	0.06	0.19	0.17
Z	0.03	0.11	0.26

QUESTION + ANSWER CHOICES

Which of the following statements gives the most fundamental reason why ornithine is unlikely to be found in proteins synthesized in vivo?

- **A.**
  There is no codon for it in the standard genetic code.
- B.
  It cannot form a peptide bond.
- C.
  It is not available in the diet.
- D.
  It has a net positive charge in aqueous solution.

SOLUTION

Solution: The correct answer is **A**.

1. **Proteins are generated during translation. In this process, mRNA triplets are read by ribosomes, and tRNA molecules provide the amino acids. Each amino acid is matched to a tRNA molecule carrying a specific anti-codon. A specific amino acid will be added only if it is recognized by its specific codon. Based on the triplet code, no codon recognizes ornithine, thus ornithine cannot integrate proteins primary structure.**
2. As ornithine is an amino acid, it is capable of forming peptide bonds.
3. As the passage indicates, ornithine is most likely provided by the diet.
4. The charge of the amino acid does not prevent it from being incorporated in the protein primary sequence.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
x
C
D
Time Spent:
4 mins 18 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1B
FC1
BIO

Question 33 of 59

Data ID: 831479
Captured: 6/25/2026, 6:24:58 PM

PASSAGE

Passage 6 (Questions 32-35)

The amino acids from which most proteins are made are called the common amino acids. Most share the general formula shown in Figure 1.

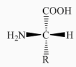

Figure 1

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$$\text{ornithine} + H_2O \rightarrow H_2NCH_2CH_2CH_2CH_2NH_2 + HCO_3^-$$

Equation 1

In the laboratory, the rate of this reaction is measured by an enzyme assay that uses ornithine in which the carboxylic acid carbon is radioactively labeled. The reaction is terminated by acidification of the reaction mixture, which converts the HCO_3^- ion to CO_2 and H_2O.

Thin-layer chromatography is an effective way of separating amino acids. Table 1 below shows R_f values for four amino acids using various solvents. "Orn" represents ornithine. "X," "Y," and "Z" represent three other amino acids.

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Orn	0.04	0.14	0.24
X	0.15	0.47	0.74
Y	0.06	0.19	0.17
Z	0.03	0.11	0.26

QUESTION + ANSWER CHOICES

Which of the following terms best describes the role of ornithine decarboxylase in the reaction shown in Equation 1?

- **A.**
  Catalyst
- B.
  Cofactor
- C.
  Substrate
- D.
  Activator

SOLUTION

Solution: The correct answer is **A**.

1. **The passage indicates that ornithine is converted to diaminobutane via ornithine decarboxylase reaction, inferring that ornithine decarboxylase is an enzyme. Enzymes are classified as catalysts in biochemical reactions.**
2. The passage infers that ornithine decarboxylase is an enzyme. Enzymes are classified as catalysts, rather than cofactors in biochemical reactions.
3. As described in the passage, ornithine decarboxylase converts ornithine to diaminobutane. This infers that ornithine decarboxylase is an enzyme rather than a substrate.
4. The information in the passage indicates that ornithine decarboxylase is involved in a biochemical reaction, inferring that it is an enzyme, rather than an activator.

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RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
38 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1A
FC1
BCH

Question 34 of 59

Data ID: 831480
Captured: 6/25/2026, 6:24:58 PM

PASSAGE

Passage 6 (Questions 32-35)

The amino acids from which most proteins are made are called the common amino acids. Most share the general formula shown in Figure 1.

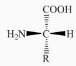

Figure 1

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QUESTION + ANSWER CHOICES

The ornithine decarboxylase reaction has been studied extensively by biomedical researchers. The most likely reason for the interest of these researchers is that the reaction is:

- A.
  a good source of ornithine.
- **B.**
  an early event in cell division.
- C.
  unique to mammals.
- D.
  one in which a chiral reactant is converted to an achiral product.

SOLUTION

Solution: The correct answer is **B.**

1. The ornithine decarboxylase reaction is an event that will result in ornithine degradation rather than being a source of ornithine.
2. **The passage states that, in mammals and other organisms, ornithine decarboxylase is active during the early events of cell division.**
3. The passage indicates that ornithine decarboxylase reaction takes place in mammals and other organisms, thus it is not unique to mammals.
4. An isomerase (epimerase and racemase), not a decarboxylase, is capable of changing the chirality of a molecule.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
39 secs
Difficulty Level:
Content & Skills:
SIRS3
CC1B
FC1
BIO

Question 35 of 59

Data ID: 831481
Captured: 6/25/2026, 6:24:59 PM

PASSAGE

Passage 6 (Questions 32-35)

The amino acids from which most proteins are made are called the common amino acids. Most share the general formula shown in Figure 1.

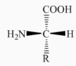

Figure 1

Amino acids can be classified on the basis of the chemical properties of their R groups. Some have hydrophobic, nonpolar R groups (e.g., valine: R = $-CH(CH_3)_2$); some have polar, uncharged R groups and are hydrophilic (e.g., serine: R = $-CH_2OH$); and others have polar, hydrophilic R groups that are acidic (e.g., aspartic acid: R = $-CH_2COOH$) or basic (e.g., lysine: R = $-CH_2CH_2CH_2CH_2NH_2$).

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Equation 1

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X	0.15	0.47	0.74
Y	0.06	0.19	0.17
Z	0.03	0.11	0.26

QUESTION + ANSWER CHOICES

The statement that the ornithine decarboxylase assay is highly specific means that it:

- A.
  requires radioactive ornithine of high specific activity.
- B.
  generates diaminobutane of high specific activity.
- **C.**
  can distinguish ornithine decarboxylase activity from the many other enzymatic reactions in a cell.
- D.
  can measure the small amount of ornithine present in a cell.

SOLUTION

Solution: The correct answer is **C**.

1. Enzyme assays are typically very specific and only measures the activity of a single chemical reaction catalyzed by a single enzyme. Onithine is an amino acid and does not possess an enzymatic specific activity.
2. Diaminobutane is a four carbon diamine and does not possess specific activity.
3. **Enzymes such as ornithine decarboxylase are highly specific both in the reactions that they catalyze and in their choice of substrates. Therefore, ornithine decarboxylase assay is very specific and only measures the activity of a single chemical reaction catalyzed by ornithine decarboxylase.**
4. Enzyme specificity relates to the type of substrate for the reaction being catalyzed and not the quantity of substrate present in the cell.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 52 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1A
FC1
BCH

Question 36 of 59

Data ID: 831503
Captured: 6/25/2026, 6:25:00 PM

PASSAGE

Passage 7 (Questions 36-39)

Many lipid-soluble toxins must be metabolized before they can be excreted from an organism. In mammalian systems, this metabolic process is often catalyzed by a type of membrane-bound heme protein known as cytochrome P-450 (*c*P-450). Many forms (isoenzymes) of *c*P-450 have been isolated, all of which contain iron with 4 coordinate bonds to nitrogen atoms in the heme group and 1 coordinate bond to a cysteine residue in the protein portion of the enzyme.

The general reaction by which *c*P-450 detoxifies many compounds is shown in Reaction 1 where RH is a lipid-soluble toxin.

$$NADPH + H^+ + O_2 + RH \xrightarrow{cP\text{-}450} NADP^+ + H_2O + ROH$$

Reaction I

*c*P-450 is inducible. That is, when an organism is challenged by a new toxin or by increased concentrations of a toxin, the organism can increase its concentration of *c*P-450 and thus metabolize and excrete the toxin more effectively.

However, *c*P-450 does not metabolize all toxins at the same rate. For example, although *c*P-450 usually metabolizes barbiturates, alcohol acts as a competitive inhibitor of barbiturate metabolism.

The role of *c*P-450 in the metabolism of toxins may be exemplified by the following case in which a chronic alcoholic died of a barbiturate overdose.

While sober, the alcoholic was having trouble falling asleep and, therefore, took the recommended dosage of sleeping pills (barbiturates). When no drowsiness was felt, the alcoholic consumed more sleeping pills, then drank an alcoholic beverage. Severe respiratory depression and death soon followed caused by an overdose of barbiturates. An autopsy revealed that if alcohol had not been present, the barbiturates would have been metabolized rapidly enough to be eliminated by the body, and death would not have occurred.

QUESTION + ANSWER CHOICES

In the example of the alcoholic in the passage, the presence of alcohol caused death because the alcohol:

- A.
  denatured the *c*P-450.
- **B.**
  inhibited the *c*P-450.
- C.
  decreased the concentration of *c*P-450.
- D.
  increased the concentration of *c*P-450.

SOLUTION

Solution: The correct answer is **B**.

1. The passage indicates that alcohol is an inhibitor of *c*P-450, not a molecule able to degrade it.
2. **Based on the passage alcohol inhibits *c*P-450, thus preventing it from metabolizing barbiturate.**
3. According to the passage alcohol does not decrease *c*P-450 concentration.
4. If the concentration of *c*P-450 was increased, the barbiturate would have been eliminated faster, avoiding the death of the individual.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✔
C
D
Time Spent:
2 mins 34 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1A
FC1
BIO

Question 37 of 59

Data ID: 831505
Captured: 6/25/2026, 6:25:00 PM

PASSAGE

Passage 7 (Questions 36-39)

Many lipid-soluble toxins must be metabolized before they can be excreted from an organism. In mammalian systems, this metabolic process is often catalyzed by a type of membrane-bound heme protein known as cytochrome P-450 (*c*P-450). Many forms (isoenzymes) of *c*P-450 have been isolated, all of which contain iron with 4 coordinate bonds to nitrogen atoms in the heme group and 1 coordinate bond to a cysteine residue in the protein portion of the enzyme.

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QUESTION + ANSWER CHOICES

Ethanol may be metabolized to acetic acid, then condensed with a coenzyme to form acetyl coenzyme A. Acetyl coenzyme A may then participate in:

- **A.**
  the Krebs (citric acid) cycle.
- B.
  glycolysis.
- C.
  electron transport.
- D.
  oxidative phosphorylation.

SOLUTION

Solution: The correct answer is **A**.

1. **Acetyl-CoA is the molecule that enters the Krebs cycle.**
2. The molecule that enters glycolysis is glucose, not Acetyl-CoA.
3. Acetyl-CoA enters the Krebs cycle, not the electron transport chain.
4. Acetyl-CoA enters the Krebs cycle, not oxidative phosphorylation. Oxidative phosphorylation is the process that follows the Krebs cycle.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
1 min 2 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1D
FC1
BIO

Question 38 of 59

Data ID: 831507
Captured: 6/25/2026, 6:25:01 PM

PASSAGE

Passage 7 (Questions 36-39)

Many lipid-soluble toxins must be metabolized before they can be excreted from an organism. In mammalian systems, this metabolic process is often catalyzed by a type of membrane-bound heme protein known as cytochrome P-450 (*c*P-450). Many forms (isoenzymes) of *c*P-450 have been isolated, all of which contain iron with 4 coordinate bonds to nitrogen atoms in the heme group and 1 coordinate bond to a cysteine residue in the protein portion of the enzyme.

The general reaction by which *c*P-450 detoxifies many compounds is shown in Reaction 1 where RH is a lipid-soluble toxin.

$$\text{NADPH} + \text{H}^+ + \text{O}_2 + \text{RH} \xrightarrow{c\text{P-450}} \text{NADP}^+ + \text{H}_2\text{O} + \text{ROH}$$

Reaction I

*c*P-450 is inducible. That is, when an organism is challenged by a new toxin or by increased concentrations of a toxin, the organism can increase its concentration of *c*P-450 and thus metabolize and excrete the toxin more effectively.

However, *c*P-450 does not metabolize all toxins at the same rate. For example, although *c*P-450 usually metabolizes barbiturates, alcohol acts as a competitive inhibitor of barbiturate metabolism.

The role of *c*P-450 in the metabolism of toxins may be exemplified by the following case in which a chronic alcoholic died of a barbiturate overdose.

While sober, the alcoholic was having trouble falling asleep and, therefore, took the recommended dosage of sleeping pills (barbiturates). When no drowsiness was felt, the alcoholic consumed more sleeping pills, then drank an alcoholic beverage. Severe respiratory depression and death soon followed caused by an overdose of barbiturates. An autopsy revealed that if alcohol had not been present, the barbiturates would have been metabolized rapidly enough to be eliminated by the body, and death would not have occurred.

QUESTION + ANSWER CHOICES

The lung cells of heavy smokers would be expected to have greatly increased concentrations of *c*P-450 and:

- A.
  DNA sequences that code for *c*P-450.
- **B.**
  mRNA sequences that code for *c*P-450.
- C.
  rRNA that process *c*P-450.
- D.
  tRNA that are specific for cysteine.

SOLUTION

Solution: The correct answer is **B**.

1. *c*P-450 DNA is present in the genome, and its levels do not change.
2. **The levels of *c*P-450 mRNA should be elevated in smokers as they are almost directly related to the levels of *c*P-450 protein produced to get rid of the toxins.**
3. The levels of ribosomal RNA (rRNA) should not change in the epithelial cells of smokers.
4. tRNA levels specific for cysteine are not related to the production of *c*P-450 protein.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
3 mins 9 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1B
FC1
BIO

Question 39 of 59

Data ID: 831511
Captured: 6/25/2026, 6:25:01 PM

PASSAGE

Passage 7 (Questions 36-39)

Many lipid-soluble toxins must be metabolized before they can be excreted from an organism. In mammalian systems, this metabolic process is often catalyzed by a type of membrane-bound heme protein known as cytochrome P-450 (*c*P-450). Many forms (isoenzymes) of *c*P-450 have been isolated, all of which contain iron with 4 coordinate bonds to nitrogen atoms in the heme group and 1 coordinate bond to a cysteine residue in the protein portion of the enzyme.

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Reaction I

*c*P-450 is inducible. That is, when an organism is challenged by a new toxin or by increased concentrations of a toxin, the organism can increase its concentration of *c*P-450 and thus metabolize and excrete the toxin more effectively.

However, *c*P-450 does not metabolize all toxins at the same rate. For example, although *c*P-450 usually metabolizes barbiturates, alcohol acts as a competitive inhibitor of barbiturate metabolism.

The role of *c*P-450 in the metabolism of toxins may be exemplified by the following case in which a chronic alcoholic died of a barbiturate overdose.

While sober, the alcoholic was having trouble falling asleep and, therefore, took the recommended dosage of sleeping pills (barbiturates). When no drowsiness was felt, the alcoholic consumed more sleeping pills, then drank an alcoholic beverage. Severe respiratory depression and death soon followed caused by an overdose of barbiturates. An autopsy revealed that if alcohol had not been present, the barbiturates would have been metabolized rapidly enough to be eliminated by the body, and death would not have occurred.

QUESTION + ANSWER CHOICES

No drowsiness was initially felt by the alcoholic because the previous abuse of alcohol had:

- A.
  denatured the *c*P-450.
- B.
  inhibited the *c*P-450.
- C.
  reduced the *c*P-450.
- **D.**
  induced the *c*P-450.

SOLUTION

Solution: The correct answer is **D**.

1. If *c*P-450 was denatured, the individuals would have felt drowsiness as the barbiturate would have had an immediate effect.
2. Inhibition of *c*P-450 would have resulted in drowsiness felt by the individual, not the opposite.
3. The passage indicates that *c*P-450 expression is inducible. Thus previous alcohol abuse will induce, rather than reduce, *c*P-450.
4. **The passage indicates that *c*P-450 is involved in barbiturate elimination and that its expression can be induced by higher levels of toxin, alcohol in this case. This suggests that higher levels of alcohol result in a rise of *c*P-450 levels and faster barbiturate elimination. As a consequence, higher doses of barbiturate will be needed to obtain drowsiness.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
x
C
D
Time Spent:
1 min 35 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1A
FC1
BIO

Question 40 of 59

Data ID: 831519
Captured: 6/25/2026, 6:25:02 PM

PASSAGE

Passage 8 (Questions 40-43)

Endothelial cells (ECs) define the lumen of capillaries and are often associated with elongated, contractile cells called *pericytes* (Figure 1).

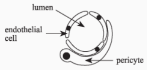


Figure 1

The following experiments were performed to test the effect of pericytes on the growth rate of ECs. In both experiments, ECs were cocultured with either pericytes, smooth-muscle cells, or fibroblasts. All cells were obtained from mammalian cell cultures.

Experiment 1: Coculture with Contact

Equal numbers of endothelial cells were placed in three separate containers (labeled A–C). Pericytes, smooth-muscle cells, and fibroblasts were *growth-arrested*—that is, treated so that they would not divide but other metabolic processes would function normally. The growth-arrested cells were mixed directly with the ECs in Containers A–C. Three matched control containers received only ECs. The number of cells in each container was counted electronically over 14 days; results are shown in Figure 2.

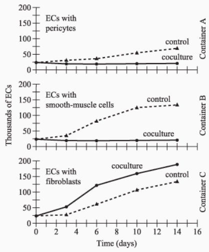


Figure 2

Experiment 2: Coculture without Contact

Equal numbers of ECs were added to two containers (D and E). Growth-arrested pericytes and smooth-muscle cells were then placed in Containers D and E, respectively; however, these cells were separated from the ECs by a semipermeable membrane. Matched control containers received only ECs. The number of cells was again counted for 14 days; results are shown in Figure 3.

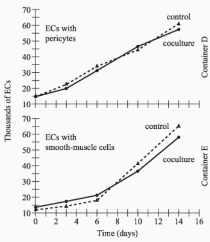


Figure 3

Figures 2 and 3 adapted from Alicia Orlidge and Patricia A. D'Amore, "Inhibition of Capillary Endothelial Cell Growth by Pericytes and Smooth Muscle Cells." ©1987 by The Rockefeller University Press.

QUESTION + ANSWER CHOICES

The pericytes used in these experiments were probably in which phase of the cell cycle?

- A.
  Telophase
- B.
  Metaphase
- C.
  Anaphase
- **D.**
  Interphase

SOLUTION

Solution: The correct answer is **D**.

1. Telophase is one of the last phases of mitosis. Cells that are in telophase are dividing.
2. Metaphase is the phase of mitosis during which the chromosomes are located in the middle of the cell plate. Cells that are in mitosis are dividing.
3. Anaphase is the phase of mitosis where sister chromatids are separated. Cells that are in anaphase are dividing.
4. **The passage indicates that pericytes and other supportive cells were growth arrested, thus they were not dividing. Cells that are in interphase are not actively dividing.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
x
C
D
Time Spent:
2 mins 6 secs
Difficulty Level:
Content & Skills:
SIRS2
CC2C
FC2
BIO

Question 41 of 59

Data ID: 831524
Captured: 6/25/2026, 6:25:03 PM

PASSAGE

Passage 8 (Questions 40-43)

Endothelial cells (ECs) define the lumen of capillaries and are often associated with elongated, contractile cells called *pericytes* (Figure 1).

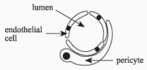


Figure 1

The following experiments were performed to test the effect of pericytes on the growth rate of ECs. In both experiments, ECs were cocultured with either pericytes, smooth-muscle cells, or fibroblasts. All cells were obtained from mammalian cell cultures.

Experiment 1: Coculture with Contact

Equal numbers of endothelial cells were placed in three separate containers (labeled A–C). Pericytes, smooth-muscle cells, and fibroblasts were *growth-arrested*—that is, treated so that they would not divide but other metabolic processes would function normally. The growth-arrested cells were mixed directly with the ECs in Containers A–C. Three matched control containers received only ECs. The number of cells in each container was counted electronically over 14 days; results are shown in Figure 2.

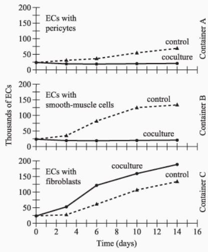


Figure 2

Experiment 2: Coculture without Contact

Equal numbers of ECs were added to two containers (D and E). Growth-arrested pericytes and smooth-muscle cells were then placed in Containers D and E, respectively; however, these cells were separated from the ECs by a semipermeable membrane. Matched control containers received only ECs. The number of cells was again counted for 14 days; results are shown in Figure 3.

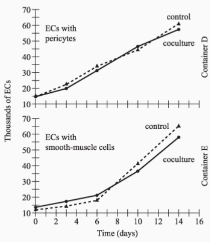


Figure 3

Figures 2 and 3 adapted from Alicia Orlidge and Patricia A. D'Amore, "Inhibition of Capillary Endothelial Cell Growth by Pericytes and Smooth Muscle Cells." ©1987 by The Rockefeller University Press.

QUESTION + ANSWER CHOICES

A student hypothesized that EC growth might be affected by the DNA from circulating erythrocytes. Is this student's hypothesis reasonable?

- A.
  No; the DNA in circulating erythrocytes is needed to help transport O_2 through the capillaries.
- **B.**
  No; circulating erythrocytes do not contain DNA.
- C.
  Yes; DNA is responsible for cell division in most cells.
- D.
  Yes; circulating erythrocytes carry DNA nutrients through the capillaries.

SOLUTION

Solution: The correct answer is **B**.

1. Mature erythrocytes do not contain DNA. It is proteins, not DNA, that help transport oxygen.
2. **Mature erythrocytes are enucleated cells that do not contain DNA.**
3. It is the levels of specific proteins, like cyclins and MPF, that regulate cell division.
4. Erythrocytes do not carry DNA, nor DNA nutrients, through the capillaries because erythrocytes are enucleated cells.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 15 secs
Difficulty Level:
Content & Skills:
SIRS2
CC3B
FC3
BIO

Question 42 of 59

Data ID: 831526
Captured: 6/25/2026, 6:25:03 PM

PASSAGE

Passage 8 (Questions 40-43)

Endothelial cells (ECs) define the lumen of capillaries and are often associated with elongated, contractile cells called *pericytes* (Figure 1).

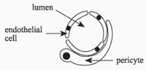


Figure 1

The following experiments were performed to test the effect of pericytes on the growth rate of ECs. In both experiments, ECs were cocultured with either pericytes, smooth-muscle cells, or fibroblasts. All cells were obtained from mammalian cell cultures.

Experiment 1: Coculture with Contact

Equal numbers of endothelial cells were placed in three separate containers (labeled A–C). Pericytes, smooth-muscle cells, and fibroblasts were *growth-arrested*—that is, treated so that they would not divide but other metabolic processes would function normally. The growth-arrested cells were mixed directly with the ECs in Containers A–C. Three matched control containers received only ECs. The number of cells in each container was counted electronically over 14 days; results are shown in Figure 2.

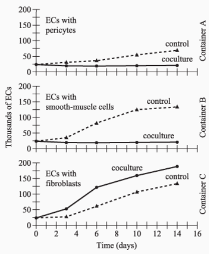


Figure 2

Experiment 2: Coculture without Contact

Equal numbers of ECs were added to two containers (D and E). Growth-arrested pericytes and smooth-muscle cells were then placed in Containers D and E, respectively; however, these cells were separated from the ECs by a semipermeable membrane. Matched control containers received only ECs. The number of cells was again counted for 14 days; results are shown in Figure 3.

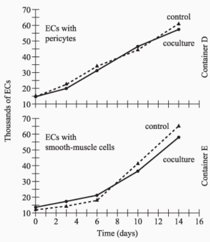


Figure 3

QUESTION + ANSWER CHOICES

Pericytes were growth-arrested in Experiment 1 so that the:

- **A.**
  pericyte growth would not interfere with the measurement of EC growth.
- B.
  ~~pericytes would not inhibit EC growth.~~
- C.
  ~~metabolic wastes of the pericytes would not interfere with the measurement of EC growth.~~
- D.
  ~~ECs would not cause the pericytes to grow~~.

SOLUTION

Solution: The correct answer is **A**.

1. **The passage indicates that endothelial cells and supportive cells were mixed together and the cell count was done electronically. This method will not allow identification of the type of dividing cell. Thus, the best way to ensure only dividing endothelial cells are counted would be to prevent other cells, in this case pericytes, from dividing.**
2. Although it might be possible that the presence of too many pericytes negatively affects the growth of endothelial cells, this is not the reason why pericyte growth was arrested.
3. It is quite unlikely that the metabolic waste might interfere with endothelial cell growth. It is most likely that the presence of more pericytes might favor endothelial cell growth.
4. The purpose of the experiment is to evaluate whether the pericytes support endothelial cell growth, and not the opposite.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
1 min 56 secs
Difficulty Level:
Content & Skills:
SIRS3
CC3B
FC3
BIO

Question 43 of 59

Data ID: 831528
Captured: 6/25/2026, 6:25:04 PM

PASSAGE

Passage 8 (Questions 40-43)

Endothelial cells (ECs) define the lumen of capillaries and are often associated with elongated, contractile cells called *pericytes* (Figure 1).

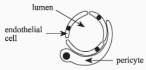


Figure 1

The following experiments were performed to test the effect of pericytes on the growth rate of ECs. In both experiments, ECs were cocultured with either pericytes, smooth-muscle cells, or fibroblasts. All cells were obtained from mammalian cell cultures.

Experiment 1: Coculture with Contact

Equal numbers of endothelial cells were placed in three separate containers (labeled A–C). Pericytes, smooth-muscle cells, and fibroblasts were *growth-arrested*—that is, treated so that they would not divide but other metabolic processes would function normally. The growth-arrested cells were mixed directly with the ECs in Containers A–C. Three matched control containers received only ECs. The number of cells in each container was counted electronically over 14 days; results are shown in Figure 2.

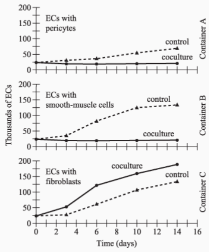


Figure 2

Experiment 2: Coculture without Contact

Equal numbers of ECs were added to two containers (D and E). Growth-arrested pericytes and smooth-muscle cells were then placed in Containers D and E, respectively; however, these cells were separated from the ECs by a semipermeable membrane. Matched control containers received only ECs. The number of cells was again counted for 14 days; results are shown in Figure 3.

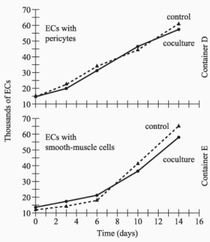


Figure 3

QUESTION + ANSWER CHOICES

Based on Figure 1 and the passage, which of the following cells are most important in the exchange of O_2 between the blood and the surrounding tissues?

- A.
  Pericytes
- **B.**
  Endothelial cells
- C.
  Smooth-muscle cells
- D.
  Fibroblasts

SOLUTION

Solution: The correct answer is **B**.

1. Pericytes are located between the endothelial cells and the matrix and they do not have direct contact with circulating blood.
2. **Endothelial cells are the cells that are in direct contact with blood and the surrounding matrix so these are the cells that play the most important role in gas exchange.**
3. Smooth-muscle cells are not in direct contact with blood, thus they are least likely to play a critical role in gas exchange between blood and the surrounding tissues.
4. Fibroblasts are supportive cells that are not in direct contact with the circulating blood, thus they are not likely to play a role in gas exchange between blood and the surrounding tissues.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
3 mins 7 secs
Difficulty Level:
Content & Skills:
SIRS4
CC3B
FC3
BIO

Question 44 of 59

Data ID: 831530
Captured: 6/25/2026, 6:25:05 PM

QUESTION + ANSWER CHOICES

The process of culturing bacteria often involves inoculation of cells on a noncellular, agar-based medium. Such a methodology would NOT result in growth of animal viruses because animal viruses:

- **A.**
  are obligate parasites.
- B.
  lack DNA.
- C.
  assimilate carbon.
- D.
  require essential vitamin supplements for growth.

SOLUTION

Solution: The correct answer is **A**.

1. **Animal viruses can only infect animal cells, not bacteria. Thus, animal viruses cannot grow in agar plates with bacteria.**
2. Some animal viruses contain DNA. The inability to culture animal viruses on agar plates is not due to the structure of their genome, but rather to their selective capability of infection.
3. Carbon assimilation is not the reason why viruses cannot be grown in bacteria seeded agar plates, but rather because animal viruses cannot infect bacteria.
4. It is the animal cells infected by the animal viruses that will require essential vitamin supplements for growth, and not the viruses per se. Animal viruses will infect only animal cells, thus, they cannot be grown in bacteria seeded agar plates.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
45 secs
Difficulty Level:
Content & Skills:
SIRS2
CC2B
FC2
BIO

Question 45 of 59

Data ID: 831531
Captured: 6/25/2026, 6:25:05 PM

QUESTION + ANSWER CHOICES

In general, telomeres are NOT important to bacterial cells because most bacterial chromosomes:

- A.
  do not replicate.
- **B.**
  are circular.
- C.
  replicate quickly and efficiently.
- D.
  are composed of single-stranded DNA.

SOLUTION

Solution: The correct answer is **B**.

1. Bacterial chromosomes replicate and their DNA copies are distributed to their daughter cells like in eukaryotic cells.
2. **The term telomere refers to a distal end of a chromosome. Telomeres contain non-coding repeats. As opposed to eukaryotic cells, bacterial DNA is constituted by one single circular molecule, thus it does not contain telomeres.**
3. Lack of telomeres is not due to the speed of replication.
4. Bacterial DNA is double stranded like the eukaryotic DNA.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

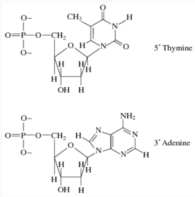
Correct
Correct Answer Your Answer
A
B
✔
C
D
Time Spent:
16 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1B
FC1
BIO

Question 46 of 59

Data ID: 831532
Captured: 6/25/2026, 6:25:06 PM

QUESTION + ANSWER CHOICES

Which of the following best describes the bond that would form between the following two nucleotides if they were located adjacent to each other as shown in a single strand of DNA?



- A.
A bond ~~between the phosphate of the thymine and the phosphate of the adenine~~
- B.
A bond between an oxygen in the thymine base and a nitrogen in the adenine base
- C.
~~A bond between the phosphate of the thymine and the sugar of the adenine~~
- **D.**
~~A bond between the phosphate of the adenine and the sugar of the thymine~~

SOLUTION

Solution: The correct answer is **D**.

1. Nucleotides are linked to one another by phosphodiester bonds between a hydroxyl group of the sugar base of the nucleotide at the 5′ (thymine) end and the phosphate group of the adjacent nucleotide at the 3′ end (adenine). In this option, thymine is placed 5′ to adenine.
2. The oxygens and nitrogens of nucleotides are not involved in bond formation between adjacent nucleotides.
3. This is the opposite of the bond that would form between a thymine at 5′ end and an adjacent adenine at 3′ end.
4. **Nucleotides are linked to one another by phosphodiester bonds between the sugar base of one nucleotide (thymine) and the phosphate group of the adjacent nucleotide (adenine) in a way that the 5′ end bears a phosphate, and the 3′ end a hydroxyl group.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
x
C
D
Time Spent:
56 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1B
FC1
BCH

Question 47 of 59

Data ID: 831534
Captured: 6/25/2026, 6:25:06 PM

QUESTION + ANSWER CHOICES

During an action potential, the movement of sodium ions into a neuron causes the neuronal membrane to do which of the following?

- A.
  Inactivate
- B.
  Hyperpolarize
- C.
  Repolarize
- **D.**
  Depolarize

SOLUTION

Solution: The correct answer is **D**.

1. The flux of sodium into the cytoplasm of the cell results in membrane depolarization. This is required for activation, not inactivation, of neuronal action potentials.
2. Hyperpolarization occurs when the membrane potential becomes more electronegative. On the contrary, the flux of sodium into the cytoplasm of the cell would result in a reduction of the membrane potential and thus depolarization instead of hyperpolarization.
3. Repolarization occurs after the membrane has been depolarized and is due to the opening of potassium channels. Sodium channels are typically inactivated during this period.
4. **As sodium enters the cytoplasm of the cell, the membrane potential will increase and the cell will depolarize.**

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
14 secs
Difficulty Level:
Content & Skills:
SIRS2
CC3A
FC3
BIO

Question 48 of 59

Data ID: 831547
Captured: 6/25/2026, 6:25:07 PM

PASSAGE

Passage 9 (Questions 48-51)

A 52-year-old female patient shows symptoms of generalized weakness, low blood pressure (80/50 mmHg), weight loss, nausea, and an extreme craving for salt. Her adrenal cortex is found to be atrophied and to contain both lymphocytes and adrenal autoantibodies. A diagnosis of primary adrenal gland insufficiency, or *Addison's disease*, is made.

Addison's disease occurs when cells of the adrenal cortex are destroyed, leaving the gland unable to secrete either glucocorticoids or mineralocorticoids. A major function of cortisol, the body's primary glucocorticoid, is to stimulate gluconeogenesis (formation of glucose from noncarbohydrate sources) in the liver by activating DNA transcription to produce liver enzymes and by mobilizing amino acids from muscle tissue. Aldosterone, the primary mineralocorticoid, maintains ionic balance by causing conservation of Na^+ and excretion of K^+.

Addison's disease may arise either as a result of tuberculosis or as an autoimmune disease. Adrenal autoantibodies are not present when tuberculosis is the cause.

To treat her Addison's disease, this patient must take both glucocorticoids and mineralocorticoids for the remainder of her life. With this medication, her life span is expected to be normal.

QUESTION + ANSWER CHOICES

The aldosterone deficiency associated with Addison's disease will cause a decrease in the serum levels of all of the following ions EXCEPT:

- A.
  ~~Na^+ ions.~~
- B.
  Cl^- ions.
- **C.**
  K^+ ions.
- D.
  HCO_3^- ions.

SOLUTION

Solution: The correct answer is **C**.

1. The passage indicates that aldosterone is responsible for sodium ion conservation in the body, thus in the absence of aldosterone, the levels of sodium ions inside the body will decrease.
2. Chloride ions movements follow the movements of sodium ions. Thus, if the levels of sodium ions decline, the levels of chloride ions will decline as well.
3. **The passage indicates that aldosterone is responsible for the elimination of potassium ions from the body. Thus, in the absence of aldosterone, the levels of potassium will increase.**
4. The passage indicates a role for corticosteroids in the regulation of sodium and potassium. The passage does not provide any information on the role of the adrenal gland on HCO_3^- levels, which most likely stay constant.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
2 mins 43 secs
Difficulty Level:
Content & Skills:
SIRS2
CC3B
FC3
BIO

Question 49 of 59

Data ID: 831551
Captured: 6/25/2026, 6:25:08 PM

PASSAGE

Passage 9 (Questions 48-51)

A 52-year-old female patient shows symptoms of generalized weakness, low blood pressure (80/50 mmHg), weight loss, nausea, and an extreme craving for salt. Her adrenal cortex is found to be atrophied and to contain both lymphocytes and adrenal autoantibodies. A diagnosis of primary adrenal gland insufficiency, or *Addison's disease*, is made.

Addison's disease occurs when cells of the adrenal cortex are destroyed, leaving the gland unable to secrete either glucocorticoids or mineralocorticoids. A major function of cortisol, the body's primary glucocorticoid, is to stimulate gluconeogenesis (formation of glucose from noncarbohydrate sources) in the liver by activating DNA transcription to produce liver enzymes and by mobilizing amino acids from muscle tissue. Aldosterone, the primary mineralocorticoid, maintains ionic balance by causing conservation of Na^+ and excretion of K^+.

Addison's disease may arise either as a result of tuberculosis or as an autoimmune disease. Adrenal autoantibodies are not present when tuberculosis is the cause.

To treat her Addison's disease, this patient must take both glucocorticoids and mineralocorticoids for the remainder of her life. With this medication, her life span is expected to be normal.

QUESTION + ANSWER CHOICES

Normally, a hypothalamic factor stimulates the release of adrenocorticotropic hormone (ACTH) from the pituitary gland. In a patient with Addison's disease, the secretion of the hypothalamic factor will:

- A.
  be lower than normal.
- **B.**
  be higher than normal.
- C.
  be unchanged.
- D.
  increase before disease onset and decrease thereafter.

SOLUTION

Solution: The correct answer is **B**.

1. Negative feedback is the mechanism that regulates the release of corticotropin releasing hormone (CRH) from the hypothalamus. In the absence or in the presence of low levels of circulating cortisol, like in Addison's disease, CRH secretion is not inhibited, thus CRH levels will be higher, not lower, than normal.
2. **Release of ACTH from the pituitary is regulated by negative feedback. In normal conditions, high levels of circulating glucose and other stressors activate the production of corticotropin-releasing hormone (CRH) from the hypothalamus. CRH will stimulate the pituitary gland to release ACTH which will trigger cortisol release from the adrenal cortex. Finally, the presence of high levels of circulating cortisol will inhibit CRH secretion (negative feedback) thus closing the loop. In Addison's disease, the circulating levels of ACTH will be higher than normal because the factor that triggers the inhibition of CRH production, high cortisol levels, is absent or low.**
3. Corticotropin releasing hormone (CRH) levels will be higher than normal, and not unchanged, in individuals suffering from Addison's disease because the levels of circulating cortisol, that normally inhibits CRH secretion, are low.
4. High levels of circulating cortisol are the stimulus to reduce corticotropin releasing hormone (CRH) secretion from the hypothalamus. In Addison's disease, the levels of cortisol are low, thus the inhibitory stimulus is absent, triggering continuous secretion and high levels of CRH. Thus, the levels of CRH are most likely low before the disease onset and high thereafter, not the opposite.

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
1 min 13 secs
Difficulty Level:
Content & Skills:
SIRS2
CC3A
FC3
BIO

Question 50 of 59

Data ID: 831565
Captured: 6/25/2026, 6:25:08 PM

PASSAGE

Passage 9 (Questions 48-51)

A 52-year-old female patient shows symptoms of generalized weakness, low blood pressure (80/50 mmHg), weight loss, nausea, and an extreme craving for salt. Her adrenal cortex is found to be atrophied and to contain both lymphocytes and adrenal autoantibodies. A diagnosis of primary adrenal gland insufficiency, or *Addison's disease*, is made.

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Addison's disease may arise either as a result of tuberculosis or as an autoimmune disease. Adrenal autoantibodies are not present when tuberculosis is the cause.

To treat her Addison's disease, this patient must take both glucocorticoids and mineralocorticoids for the remainder of her life. With this medication, her life span is expected to be normal.

QUESTION + ANSWER CHOICES

If a patient with Addison's disease is given too high a replacement dose of glucocorticoids, the effect over time will be an increase in:

- A.
  muscle mass.
- **B.**
  muscle weakness.
- C.
  white blood cell count.
- D.
  heart rate.

SOLUTION

Solution: The correct answer is **B**.

1. One of the effects of glucocorticoids is to increase protein degradation in various tissues, including muscles. Thus, high levels of circulating glucocorticoids will most likely reduce, rather than increase, muscle mass.
2. **High levels of circulating glucocorticoids will increase protein degradation in various tissues, muscles included. A direct consequence of protein degradation in muscles is muscle weakness.**
3. Glucocorticoids suppress immune function, which is mediated by white blood cells. Therefore, the activity of white blood cells will decrease, not increase, in response to excess glucocorticoid exposure.
4. Heart rate will be decreased in Addison's affected individuals treated with high doses of glucocorticoids, but these individuals will suffer from hypertension due to increased systemic vascular resistance, increased extracellular volume, and increased cardiac contractility.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
x
Time Spent:
42 secs
Difficulty Level:
Content & Skills:
SIRS2
CC3B
FC3
BIO

Question 51 of 59

Data ID: 831568
Captured: 6/25/2026, 6:25:09 PM

PASSAGE

Passage 9 (Questions 48-51)

A 52-year-old female patient shows symptoms of generalized weakness, low blood pressure (80/50 mmHg), weight loss, nausea, and an extreme craving for salt. Her adrenal cortex is found to be atrophied and to contain both lymphocytes and adrenal autoantibodies. A diagnosis of primary adrenal gland insufficiency, or *Addison's disease*, is made.

Addison's disease occurs when cells of the adrenal cortex are destroyed, leaving the gland unable to secrete either glucocorticoids or mineralocorticoids. A major function of cortisol, the body's primary glucocorticoid, is to stimulate gluconeogenesis (formation of glucose from noncarbohydrate sources) in the liver by activating DNA transcription to produce liver enzymes and by mobilizing amino acids from muscle tissue. Aldosterone, the primary mineralocorticoid, maintains ionic balance by causing conservation of Na^+ and excretion of K^+.

Addison's disease may arise either as a result of tuberculosis or as an autoimmune disease. Adrenal autoantibodies are not present when tuberculosis is the cause.

To treat her Addison's disease, this patient must take both glucocorticoids and mineralocorticoids for the remainder of her life. With this medication, her life span is expected to be normal.

QUESTION + ANSWER CHOICES

The most rapid rate of gluconeogenesis will most likely occur in the body when:

- A.
  blood glucose levels are high.
- B.
  cortisol release is inhibited.
- **C.**
  the body's stores of carbohydrates are low.
- D.
  the body's stores of proteins are low.

SOLUTION

Solution: The correct answer is **C**.

1. Gluconeogenesis becomes active when the body needs more glucose to generate energy. Therefore, gluconeogenesis will most likely occur when blood glucose levels are low, not high.
2. Gluconeogenesis becomes active when the body needs energy, so high cortisol levels increase gluconeogenesis. Therefore, gluconeogenesis is most likely downregulated when cortisol release is inhibited.
3. **Gluconeogenesis is the pathway for the synthesis of glucose (a carbohydrate) from other metabolic compounds such as lipids and amino acids. Therefore, this pathway is activated when the body's stores of carbohydrates are low.**
4. Gluconeogenesis occurs when blood glucose levels, rather than protein storage, is low.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✓
D
Time Spent:
1 min 10 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1D
FC1
BCH

Question 52 of 59

Data ID: 831585
Captured: 6/25/2026, 6:25:09 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION + ANSWER CHOICES

Glucose transporter proteins in the liver do not require the presence of insulin to facilitate the uptake of glucose. However, insulin does stimulate the first step in the glycolytic pathway within the liver. Therefore, in liver cells, insulin most likely:

- A.
  ~~hinders glucose uptake by increasing the cellular concentration of glucose.~~
- **B.**
  aids glucose uptake by decreasing the cellular concentration of glucose.
- C.
  hinders glucose uptake by using the ATP needed by the glucose transporter proteins.
- D.
  aids glucose uptake by providing the ATP needed by the glucose transporter proteins.

SOLUTION

Solution: The correct answer is **B**.

1. Based on the passage, insulin is involved in the first step of the glycolytic pathway. Therefore, it promotes, rather than hinders, glucose uptake.
2. **The question states that insulin stimulates the first step in the glycolytic pathway in the liver. This will result in a decrease of the cellular concentration of glucose. To compensate for the low cellular glucose concentration, glucose uptake is increased.**
3. The question states that insulin stimulates the first step in the glycolytic pathway in the liver and depletion of intracellular glucose levels. Therefore, insulin will promote, rather than hinder, glucose uptake. This process occurs because of the decrease in the cellular concentration of glucose, rather than using the ATP needed by the glucose transporter proteins.
4. Based on the question, insulin stimulates the first step in the glycolytic pathway within the liver. This infers that insulin promotes glucose uptake via depletion of intracellular glucose, rather than providing the ATP needed by the glucose transporter proteins.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
x
Time Spent:
2 mins 45 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1D
FC1
BCH

Question 53 of 59

Data ID: 831588
Captured: 6/25/2026, 6:25:10 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION + ANSWER CHOICES

Exercise promotes the insulin-independent uptake of glucose in working skeletal muscles. Given this, regular exercise would most likely reduce blood glucose levels in patients with which type(s) of diabetes?

- A.
  Type 1 only
- B.
  Type 2 only
- **C.**
  Both Type 1 and Type 2
- D.
  Neither Type 1 nor Type 2

SOLUTION

Solution: The correct answer is **C**.

1. As working skeletal muscles are insulin independent, exercise will reduce the levels of insulin in both types of diabetes, not just Type 1.
2. The item indicates that working skeletal muscles use glucose independently from the availability of insulin. Thus, exercise will reduce circulating glucose levels in both types of diabetes, not just Type 2.
3. **The item indicates that working skeletal muscles use glucose in an insulin-independent manner. Thus, exercise will be able to reduce the levels of glucose in both types of diabetes.**
4. Working skeletal muscle can use glucose independently from insulin availability, thus the levels of circulating glucose will be reduced in both types of diabetes during exercise.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
✔
D
Time Spent:
33 secs
Difficulty Level:
Content & Skills:
SIRS2
CC3B
FC3
BIO

Question 54 of 59

Data ID: 831592
Captured: 6/25/2026, 6:25:10 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION + ANSWER CHOICES

During the production of insulin, the translated polypeptide is cleaved into the mature form and secreted from the cell. The cleavage most likely takes place in which of the following locations?

- A.
  Nucleus
- B.
  Ribosomes
- **C.**
  Endomembrane system
- D.
  Cytoplasm

SOLUTION

Solution: The correct answer is **C**.

1. Insulin is not a transcription factor and its polypeptide is most likely not present in the nucleus.
2. Insulin polypeptide is translated, not cleaved, in the ribosomes.
3. **The endomembrane system is the portion of the cells that is in charge of modifying proteins that will be secreted. Thus, it is most likely that insulin cleavage will occur in the endomembrane system.**
4. Insulin cleavage is most likely to occur in the endomembrane system, rather than the cytoplasm, because insulin is a secreted protein.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
D
x
Time Spent:
1 min 6 secs
Difficulty Level:
Content & Skills:
SIRS1
CC2A
FC2
BIO

Question 55 of 59

Data ID: 831596
Captured: 6/25/2026, 6:25:11 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION + ANSWER CHOICES

Despite the effects of diabetes, the brains of diabetic patients still receive adequate nourishment. This is most likely because the brain uses:

- A.
  less glucose than do other body tissues.
- **B.**
  insulin-independent transporters for the uptake of glucose.
- C.
  fatty acids for energy instead of glucose.
- D.
  insulin-dependent transporters for the uptake of glucose.

SOLUTION

Solution: The correct answer is **B**.

1. The brain is the main consumer of glucose in the body.
2. **The brain cells contain glucose transporters that are insulin independent.**
3. Brain cells are unable to use fatty acids for energy.
4. If the brain cells had insulin-dependent transporters, they will not be functioning correctly in individuals affected by diabetes.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
✓
C
D
Time Spent:
49 secs
Difficulty Level:
Content & Skills:
SIRS2
CC3A
FC3
BIO

Question 56 of 59

Data ID: 831602
Captured: 6/25/2026, 6:25:12 PM

PASSAGE

Passage 10 (Questions 52-56)

A 40-year-old patient developed a constant thirst and a frequent need to urinate. The attending physician suspected the patient had diabetes because in addition to these symptoms, the patient was overweight and had a family history of diabetes, which both are risk factors for the disease.

Diabetes mellitus is a condition in which the body cannot produce or is unable to effectively respond to insulin, a peptide hormone made in pancreatic β cells. Normally, insulin promotes facilitated diffusion of glucose into cells from the bloodstream by inducing the translocation of glucose transporter proteins to the plasma membrane. It also helps regulate the metabolism of fatty and amino acids.

In diabetes, the ability of many types of cells to take up glucose is compromised, leading to elevated blood glucose levels. Many tissues then metabolize fatty acids as an alternative energy source, which can lead to *metabolic acidosis*, a condition of low blood pH. Diabetes can also cause protein degradation, resulting in the release of excess amino acids that can be used as substrates for gluconeogenesis, which further increases blood glucose levels. There are two main types of diabetes. Many cases of Type 1 diabetes, in which the body is unable to produce insulin, are caused by an autoimmune response directed against pancreatic β cells. While the body often produces normal-to-elevated amounts of insulin in Type 2 diabetes, its ability to respond to insulin is compromised.

QUESTION + ANSWER CHOICES

Based on the passage, which of the following is LEAST likely to be a symptom of diabetes mellitus?

- **A.**
  Loss of appetite
- ~~B.~~
  ~~Sweet-tasting urine~~
- C.
  Unexplained weight loss
- D.
  ~~Feelings of fatigue~~

SOLUTION

Solution: The correct answer is **A**.

1. **Based on the passage, diabetes affected individuals will use proteins and lipids as a source of glucose, thus increasing, rather than decreasing, their appetite.**
2. High levels of glucose are a symptom of diabetes. This is due to the fact that kidneys can reabsorb glucose up to 320 mg/min (tubular maximum). The levels of glucose in the filtrate that are in excess will pass into urine, leading to a sweet taste.
3. Unexplained weight loss might occur as the body uses proteins and lipids as a source of energy.
4. Feeling of fatigue might occur as the body uses amino acids and lipids as a source of energy.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
X
D
Time Spent:
1 min 19 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1D
FC1
BIO

Question 57 of 59

Data ID: 831609
Captured: 6/25/2026, 6:25:12 PM

QUESTION + ANSWER CHOICES

A certain bacterium was cultured for several generations in medium containing ^{15}N, transferred to medium containing ^{14}N, and allowed to complete two rounds of cell division. Given that the bacterium's genome mass is 5.4 fg when grown in ^{14}N media and 5.5 fg when grown in ^{15}N medium, individual bacteria with which of the following genome masses would most likely be isolated from this culture?

- A.
  5.4 fg only
- **B.**
  5.4 fg and 5.45 fg
- C.
  5.4 fg and 5.5 fg
- D.
  5.45 fg only

SOLUTION

Solution: The correct answer is **B**.

1. This would be the genome mass if the bacterium was cultured only in ^{14}N.
2. **DNA replication occurs in a semiconservative way and, after two rounds of bacterial growth, there will be bacteria containing only ^{14}N and bacteria containing the same amount of ^{15}N and ^{14}N. Thus, the genome mass will be 5.4 fg for the bacteria that only contain ^{14}N, and 5.45 fg for the bacteria that contain equal levels of ^{14}N and ^{15}N.**
3. This would be the genome mass if the bacteria were cultured only with ^{15}N or ^{14}N separately. The two rounds of division with ^{14}N will generate bacteria that contain ^{14}N and ^{15}N, as well as bacteria that only contain ^{14}N.
4. Based on the semiconservative method of DNA replication, after two rounds of cell division some bacteria will only contain ^{14}N (genome mass 5.4 fg) and some others will contain a mix of ^{14}N and ^{15}N (genome mass 5.45 fg), thus the genome mass of the entire culture cannot be only 5.45 fg.

Khan Academy Lessons

RESULT / TIMING / TAGS

Incorrect
Correct Answer Your Answer
A
B
C
x
D
Time Spent:
2 mins 23 secs
Difficulty Level:
Content & Skills:
SIRS2
CC2B
FC2
BIO

Question 58 of 59

Data ID: 831616
Captured: 6/25/2026, 6:25:13 PM

QUESTION + ANSWER CHOICES

Assume that *K* and *M* are two unlinked genes that affect hearing. The dominant *K* allele is necessary for hearing, and the dominant *M* allele causes deafness regardless of the other genes present. Given this, what fraction of the offspring of parents with the genotypes *KkMm* and *Kkmm* will most likely be deaf?

- A.
  1/4
- B.
  3/8
- C.
  1/2
- **D.**
  5/8

SOLUTION

Solution: The correct answer is **D**.

1. This would be correct if both individuals were heterozygous for *K* and homozygous recessive for *M*.
2. There are no possible genotype combination that will result in this ratio.
3. This would be correct if one parent was *Kkmm* and the other *kkmm*, or if one parent was *KkMm* and the other was *KKmm*.
4. **Based on the possible allele combination, out of 16 possible combinations, there will be ten individuals that will be hearing impaired, carrying either one allele *M* or the two *kk*. Thus 10/16 = 5/8 individuals.**

Khan Academy Lessons

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
B
C
D
✓
Time Spent:
3 mins 22 secs
Difficulty Level:
Content & Skills:
SIRS2
CC1C
FC1
BIO

Question 59 of 59

Data ID: 831618
Captured: 6/25/2026, 6:25:13 PM

QUESTION + ANSWER CHOICES

An RNA molecule has 1500 bases. What is the maximum number of amino acids it can encode?

- **A.**
  500
- B.
  1000
- C.
  1500
- D.
  4500

SOLUTION

Solution: The correct answer is **A**.

1. **Each amino acid is coded by a triplet of nucleobases. Thus, $\frac{1500}{3} = 500$ amino acids.**
2. This is the number of amino acids coded by a sequence of 3000 bases, not 1500 bases.
3. This is the number of amino acids coded by a sequence of 4500 bases, not 1500 bases.
4. This is the number of amino acids coded by a sequence of 13500 bases, not 1500 bases.

Khan Academy Lessons

For more help on the skills, concepts, or content assessed by this question, see the Khan

RESULT / TIMING / TAGS

Correct
Correct Answer Your Answer
A
✓
B
C
D
Time Spent:
30 secs
Difficulty Level:
Content & Skills:
SIRS1
CC1B
FC1
BIO